

CISCO NETWORKING BASICS

MODULE 1 — COMMUNICATION IN A CONNECTED WORLD

🔗 Networking Fundamentals

🔗 Cisco Skills for All

⇒ Handwritten Notes → Digital

COURSE

Networking Basics

MODULE

1 of Course

STARTED

05 / 06 / 2026

FORMAT

Notes + Quiz + Test

0 Course Introduction

The Networking Basics course teaches the **fundamentals of networking** by covering the basic concepts and skills needed to set up and manage your small office or home office (SOHO) network.

1 Network Types

The internet is not owned by any individual or group. It is a worldwide collection of interconnected networks (internetwork, or "internet" for short), cooperating with each other to exchange information using common standards — through telephone wires, fiber-optic cables, wireless transmission, and satellite links.

Networks exist at different scales:

- **Small home networks** — connect a few computers to each other and to the internet.
- **Small office / home office (SOHO)** — allows computers in a home office or remote office to connect to a corporate network, or access shared/centralized resources.
- **Medium to large networks** — such as those used by corporations and schools, with hundreds or thousands of interconnected hosts across many locations.
- **Worldwide networks** — the internet itself: a network of networks connecting hundreds of millions of computers globally.

2 Connected Devices

Mobile devices connected to the internet daily: smartphones, tablets, smartwatches, and smart glasses — all connect and communicate through the internet.

Connected home devices: security systems, appliances, smart TV, and gaming console.

Connected outside the home: smart cars, RFID tags, sensors and actuators, and even medical devices.

3 Data Transmission — Bits & ASCII

Computers and networks only work with **binary digits — zero and one**. It can be difficult to imagine that all of your data is stored and transmitted as a series of bits. Each bit can only have one of two possible values, **0 or 1**. The term "bit" is an abbreviation of *binary digit* and represents the smallest piece of data. Humans interpret words and pictures; computers interpret only patterns of bits.

ASCII (American Standard Code for Information Interchange): with ASCII, each character is represented by eight bits.

Type	Character	Binary (8-bit)
Capital letter	A	01000001
Number	9	00111001
Special character	#	00100011

Each group of eight bits (such as the representation of letters and numbers) is known as a **byte**.

4 Common Methods of Signal Transmission

After data is transformed into a series of bits, it must be converted into signals that can be sent across the network media to its destination. Media refers to the physical medium on which the signals are transmitted.

#	Method	How It Works
1	Electrical Signals	Achieved by representing data as electrical pulses on copper wire.
2	Optical Signals	Achieved by converting electrical signals into light pulses.
3	Wireless Signals	Achieved by using infrared, microwave, or radio waves through the air.

5 Bandwidth and Throughput

Bandwidth is the capacity of a medium to carry data. Digital bandwidth measures the amount of data that can flow from one place to another in a given amount of time. Bandwidth is typically measured in the number of bits that (theoretically) can be sent across the media in a second. Streaming a movie or playing a multiplayer game requires reliable, fast connections. To support these "high bandwidth" applications, networks have to be capable of transmitting and receiving bits at a very high rate. Different physical media support the transfer of bits at different speeds — physical media properties, current technologies, and the laws of physics all play a role in determining available bandwidth.

Unit of Bandwidth	Abbreviation	Equivalence
Bits per Second	bps	1 bps = fundamental unit of bandwidth
Kilobits per Second	Kbps	1 Kbps = 1,000 bps = 10 ³ bps
Megabits per Second	Mbps	1 Mbps = 1,000,000 bps = 10 ⁶ bps
Gigabits per Second	Gbps	1 Gbps = 1,000,000,000 bps = 10 ⁹ bps
Terabits per Second	Tbps	1 Tbps = 1,000,000,000,000 bps = 10 ¹² bps

Throughput — like bandwidth, throughput is the measure of the transfer of bits across the media over a given period of time. However, due to a number of factors, throughput does not usually match the specified bandwidth. Factors influencing throughput include:

- The amount of data being sent and received over the connection.
- The type of data being transmitted.
- The latency created by the number of network devices encountered between source and destination.

Latency refers to the amount of time, including delays, for data to travel from one given point to another.

Throughput measurements do not take into account the validity or usefulness of the bits being transmitted and received. Many messages received through the network are not destined for specific user applications — an example would be network control messages that regulate traffic and correct errors.

In an internetwork or network with multiple segments, throughput cannot be faster than the slowest link of the path from the sending device to the receiving device. Even if all or most of the segments have high bandwidth, it will only take one segment in the path with lower bandwidth to create a slowdown of the throughput for the entire network. There are many online speed tests that can reveal the throughput of an internet connection.

6 Personal Data — Types

The following categories are used to classify types of personal data:

Type	Description
Volunteered data	Created and explicitly shared by individuals, such as social network profiles. This type of data might include video files, pictures, text, or audio files.
Observed data	Captured by recording the actions of individuals, such as location data when using cell phones.
Inferred data	Data such as a credit score, which is based on analysis of volunteered or observed data.

7 In-Chapter Practice Questions

Q1. What type of network allows computers in a home office or a remote office to connect to a corporate network?

Ans — Small office / home office (SOHO) network.

Q2. What can be placed in or on a package so that it can be tracked?

Ans — RFID tag.

Q3. When using your credit card to make purchases at several places, the credit card company can use this information to learn about your location and preferences. This type of personal data is known as?

Ans — Inferred data.

Q4. Which of the following methods of signal transmission uses frequencies or pulses of light?

Ans — Optical signals.

Q5. A basic unit of information that represents one of two discrete states is known as a?

Ans — Bit.

Q6. The capacity of the medium to carry data is known as?

Ans — Bandwidth.

Q7. What measurement includes any latency encountered during data transmission?

Ans — Throughput.

Q8. What measurement is used to indicate thousands of bits per second?

Ans — Kbps.

8 Module Test — Questions & Answers

Q1. What is the internet?

Ans — A network of networks.

Q2. What is an example of a binary value from everyday life?

Ans — A simple light switch.

Q3. Which category of network components includes wires and cables used in a wired network?

Ans — Media.

Q4. What type of device is able to create physical movement?

Ans — Actuator.

Q5. What are three options for signal transmission on a network?

Ans — Light pulses, radio waves, electrical pulses.

Q6. Who owns the internet?

Ans — No one person or group.

Q7. Which type of connected device is placed on objects to track and monitor them?

Ans — RFID tags.

Q8. A byte consists of how many bits?

Ans — 8 bits.

Q9. Which two numbers are possible values of a bit?

Ans — 0 and 1.

Q10. What measurement is used to indicate thousands of bits per second?

Ans — Kbps.

Q11. What type of network must a home user access in order to do online shopping?

Ans — The internet.

CISCO NETWORKING BASICS

MODULE 2 — NETWORK COMPONENTS, TYPES AND CONNECTIONS

🔗 Networking Fundamentals

🔗 Cisco Skills for All

⇒ Handwritten Notes → Digital

COURSE

Networking Basics

MODULE

2 of Course

STARTED

04 / 08 / 2026

FORMAT

Notes + Quiz + Test

0 Module Introduction

Module 2 builds on the fundamentals from Module 1 and looks at how devices actually **participate in and connect to a network**. It covers the roles computers play (client vs. server), the two major ways networks are designed (peer-to-peer vs. client/server), the physical components that make up a network's infrastructure, and the different ways a home or small office can connect to the internet through an ISP.

1 Clients and Servers

All computers connected to a network that participate directly in network communication are classified as **hosts**. Hosts can send and receive messages on the network. In modern networks, computer hosts can act as a **client**, a **server**, or both — the software installed on the computer determines which role it plays.

- **Server** — a computer that has software installed which allows it to provide information, such as email or web pages, to other devices on the network.
- **Client** — a computer that has software installed which allows it to request and display the information obtained from the server.

Example: A printer used to share data is called **print sharing**. A file used to share data packets is called **file sharing**.

Client and server software usually run on separate computers, but it is also possible for one computer to run both client and server software at the same time.

2 Peer-to-Peer (P2P) Networking — Advantages

In a peer-to-peer (P2P) network, two or more computers are connected and can share resources (such as files or printers) without a dedicated server. Every device can act as both a client and a server.

Advantages of peer-to-peer networking:

- Easy to set up.
- Less complex.
- Lower cost, because dedicated network devices and servers may not be required.
- Can be used for simple tasks such as transferring files and sharing printers.

3 Peer-to-Peer (P2P) Networking — Disadvantages & Hybrid Systems

Disadvantages of peer-to-peer networking:

- No centralized administration.
- Not as secure.
- Not scalable.
- All devices may act as both clients and servers, which can slow down their performance.

Some P2P applications use a hybrid system where client-and-server communication is used to reference the location of resources, but the actual transfer of data still happens directly between clients. For example, in an instant-messaging network, both clients can simultaneously send and receive messages once the network has connected them:



Both clients can simultaneously send and receive messages (hybrid P2P instant messaging).

4 Network Infrastructure & Components

There are symbols that represent various types of networking equipment. The **network infrastructure** is the platform that supports the network — it provides the stable and reliable channel over which our communications can occur. The network infrastructure contains three categories of hardware components:

Category	Examples
End Devices	Desktop computer, laptop, printer, IP phone, wireless tablet, telepresence endpoint.
Intermediary Devices	Wireless router, LAN switch, router, multilayer switch, firewall appliance.
Network Media	Wireless media, LAN media, WAN media.

Hardware is often the visible component of the network platform, such as a laptop, PC, switch, router, or wireless access point — or the cabling used to connect the devices. Some components, such as wireless media, are not visible.

End devices (hosts) form the interface between users and the underlying communication network. Some examples of end devices include:

- Computers (workstations, laptops, file servers, web servers).
- Network printers.
- Telephones and tele-conferencing equipment.
- Security cameras.
- Mobile devices (such as smartphones, tablets, PDAs, and wireless debit/credit card readers and barcode scanners).

5 Internet Service Provider (ISP) Connectivity Options

An **ISP** provides the link between the home network and the internet. An ISP can be a local cable provider, a landline telephone service provider, the cellular network that provides your smartphone service, or an independent provider who leases bandwidth on the physical network infrastructure of another company.

Each ISP connects to other ISPs to form a network of links that interconnects users all over the world. ISPs are connected in a **hierarchical manner** that ensures internet traffic generally takes the shortest path from source to destination.

Common ISP services include:

- FTP hosting.
- Applications and media hosting.
- Technical support.
- Voice over IP (VoIP).
- POP internet access.
- Equipment co-location.
- Web hosting.

The interconnection of ISPs that forms the backbone of the internet is a complex web of fiber-optic cables, expensive networking switches, and routers that direct the flow of information between source and destination hosts.

Common connection options:

User Type	Connection Options
Home user	DSL, cable, cellular, satellite, dial-up telephone
Teleworker	DSL, cable, cellular, satellite, dial-up telephone
Small office	DSL, cable, cellular, satellite, dial-up telephone

For a home user, connecting to the ISP is usually a fairly uncomplicated process — it consists of using a wired or wireless integrated router to connect to the ISP. The router includes a switch to connect wired hosts, and it also provides client IP addressing information and security for inside hosts. The two most common methods are **cable** and **DSL**. Other options include **cellular**, **satellite**, and **dial-up telephone**.

6 Network Diagrams — Client/Server & ISP Models

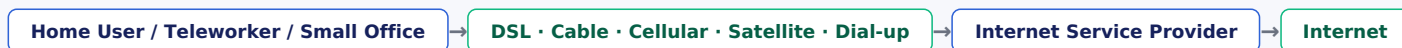
Client & Server model:



Satellite ISP workflow:



Home / Teleworker / Small Office → ISP connection paths:



Multiple PCs on a LAN reach the ISP through an integrated router; a single PC can connect directly through a modem.

7 In-Chapter Practice Questions

Q1. A computer that has software installed to provide information such as email or webpages to other devices is known as?

Ans — Server.

Q2. A smartphone uses web browser software to request and display a web page. The smartphone is considered what type of computer?

Ans — Client.

Q3. A network where two computers are communicating with each other as both a client and a server is known as a?

Ans — Peer-to-peer network.

Q4. Marjani is on a farm several kilometers outside of Msolwa, Tanzania. What device is she most likely to use to connect to the internet?

Ans — Smartphone and wireless tablet.

Q5. Eilert recently got a job with a computer support services company in his hometown of Falun, Sweden. A customer wants their home network connected to the internet, and they only have a cable modem. What intermediary device would Eilert most likely take with him to the job?

Ans — Wireless router.

Q6. Rosalia works as a community health worker in Rio Claro, Brazil. She needs internet access to maintain patient records and provide video-conferencing with a doctor when needed. What end device and media type does she most likely use?

Ans — Tablet and wireless media.

Q1. What type of network is defined by two computers that can both send and receive requests for resources?

Ans — Peer-to-peer.

Explanation: The simplest form of peer-to-peer networking consists of two computers directly connected to each other through a wired or wireless connection.

Q2. What are two functions of end devices on a network?

Ans — (1) They originate the data that flows through the network. (2) They are the interface between humans and the communication network.

Explanation: End devices originate the data that flows through the network. Intermediary devices direct data over alternate paths in the event of link failures, and filter the flow of data to enhance security. Network media provide the channel through which network messages travel.

Q3. A home user is looking for an ISP connection that provides high-speed digital transmission over regular phone lines. What ISP connection type should be used?

Ans — DSL (Digital Subscriber Line).

Explanation: DSL is the best technology to use over existing phone lines. Many ISPs have a lookup on their website where you can enter your phone number to see if they can offer you the service.

Q4. What type of internet connection would be best for a residence in a remote area without mobile phone coverage or wired connectivity?

Ans — Satellite.

Q5. Which term correctly describes the function of an ISP?

Ans — Responsible for providing the link between a private network and the internet.

Q6. Which device is an intermediary device?

Ans — Firewall.

Q7. Which scenario describes a peer-to-peer network?

Ans — A user has shared a printer attached to their workstation.

Q8. Which term is used to describe a network device whose primary function is to provide information to other devices?

Ans — Server.

Q9. What is an advantage of the peer-to-peer network model?

Ans — Easy to set up.

Q10. What is a characteristic of a peer-to-peer application?

Ans — Each device using the application provides a user interface and runs a background service.

CISCO NETWORKING BASICS

MODULE 3 — WIRELESS AND MOBILE NETWORKS

Networking Fundamentals

Cisco Skills for All

Handwritten Notes → Digital

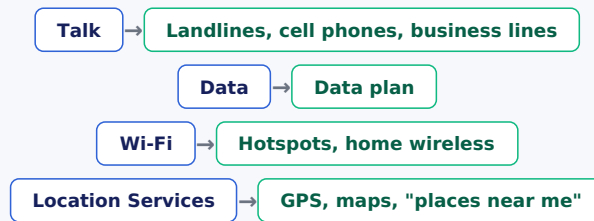
COURSE
Networking BasicsMODULE
3 of CourseSTARTED
04 / 08 / 2026FORMAT
Notes + Test

0 Module Introduction

Module 3 looks at how **wireless and mobile networks** keep us connected on the move — from the cellular towers that carry phone calls and data, to the short-range technologies (Wi-Fi, Bluetooth, NFC) that connect our everyday devices, and the basic habits needed to keep a wireless connection secure.

1 Wireless Networks & GSM

Wireless networks transmit signals from **tower to tower**. This type of network is called a **GSM network** (Global System for Mobile communications).



What a GSM/mobile network typically carries.

2 Mobile Data & Short-Range Wireless Tech

Mobile phones use **radio waves** to send signals through the GSM network. **3G, 4G, and 5G** are the generations used for progressively faster data transmission.

- **Wi-Fi** — connects a phone to the internet / a local network.
- **Bluetooth** — connects nearby devices to each other.
- **NFC (Near Field Communication)** — works only within a few centimeters.

3 Protecting Wi-Fi on Mobile Devices

To keep a mobile device's wireless connection safe:

- Never send a password in plaintext.
- Use a **VPN** for sensitive data.
- Always use **WPA2 encryption**.

4 Bluetooth Pairing

During Bluetooth pairing, a device goes into **discoverable mode** and shares its name, class, and services. A **PIN** is used to authenticate the connection.

5 Module Test — Questions & Answers

Q1. What term is used to describe connecting a mobile device to another mobile device or computer to share a network connection?

Ans — Tethering.

Q2. Which wireless technology uses a device-pairing process to communicate, and operates over short distances of up to 100 meters?

Ans — Bluetooth.

Q3. What technology enables a cell phone to be used as a hands-free device?

Ans — Bluetooth.

Q4. A student purchased a new Wi-Fi-enabled tablet computer. What is required to connect this device to the internet?

Ans — A wireless LAN (Local Area Network).

Q5. Which technology allows a mobile device to establish wireless communication with another mobile device by touching them together?

Ans — NFC (Near Field Communication).

Q6. Which technology will allow a mobile device to share an internet connection with other devices via tethering?

Ans — Bluetooth.

Q7. What are two methods typically used on a mobile device to provide internet connectivity?

Ans — Cellular, Wi-Fi.

Q8. Which wireless technology allows a customer to connect to a payment system with a smartphone?

Ans — NFC (Near Field Communication).

Q9. A salesperson is using a smartphone map application to locate a business. What wireless technology enables the smartphone to receive satellite geolocation information for the map application?

Ans — GPS (Global Positioning System).

Q10. What two pieces of information are required when manually connecting either an iOS device or an Android device to a secure wireless network?

Ans — The SSID. The password.

Q11. Which wireless technology can be used to connect wireless headphones to a computer?

Ans — Bluetooth.

CISCO NETWORKING BASICS

MODULE 4 — BUILD A HOME NETWORK

Networking Fundamentals

Cisco Skills for All

Handwritten Notes → Digital

COURSE

Networking Basics

MODULE

4 of Course

STARTED

04 / 08 / 2026

FORMAT

Notes + Quiz + Test

0 Module Introduction

Module 4 covers how to actually **build and configure a home network** — the two parts of a home network (public and private), the router that ties them together, the wireless technologies and cabling involved, and the step-by-step process of setting up and securing a home router.

1 Home Network Basics

1. A home network = 2 networks: **public** (from the ISP) + **private** (LAN).
2. The **router** connects to the internet and supports both wired and wireless devices.
3. **Advantages of wireless LAN:** mobility, scalability, flexibility, cost savings, reduced installation time, and reliable operation even in harsh environments.
4. **Devices that connect:** desktop computers, gaming systems, smart TVs, printers, scanners, security cameras, climate control devices.

2 Router Ports & Wireless Access Point

5. A router has two types of ports: **Ethernet ports** (for LAN devices) and an **internet port** (for the ISP).
6. The **wireless access point** is built into the router.

Diagram — Home Wireless Local Area Network (WLAN):



3 Quick Check

Q1. A home router typically only provides wired access to the network. You have to buy a separate device for wireless access?

Ans — The statement is false.

Q2. Which of the following is used to connect a wired device to the internal switch of the home router?

Ans — Ethernet port.

4 Network Technologies — Wireless & Wired

1. **Wireless** uses electromagnetic waves to carry data between devices.
2. Home networks use **unlicensed bands**: 2.4 GHz and 5 GHz.
3. **Bluetooth** → uses only the 2.4 GHz band.
4. **802.11 (Wi-Fi)** → uses both 2.4 GHz and 5 GHz, transmits at higher power = greater range and improved throughput vs. Bluetooth.
5. **Wired** → the Ethernet protocol is most common; it uses UTP cable.
6. **Cat 5e** is the most common wiring — it has 4 twisted pairs (which reduces electrical interference).
7. **Powerline technology** also exists — it provides wired connectivity without needing new wiring.

5 Electromagnetic Spectrum & Frequency Notes

Where wireless technologies exist on the electromagnetic spectrum (low to high frequency):

Band	Typical Use
Extremely Low / Very Low / Low	Audio
Medium	AM broadcast
High	Short-wave radio
Very High	FM broadcast, Television
Ultra High	Television, Cellular (890 MHz), NPCS (930 MHz)
Super High / Infrared	Infrared wireless LAN

Ultra High: 902–928 MHz cordless phones.

Super High: 5.725–5.850 GHz, IEEE 802.11 a/n/ac/ad.

Ultra High + Super High: 2.400–2.4835 GHz, IEEE 802.11 b/g/n/ad.

Q1. Certain areas of the electromagnetic spectrum can be used without a permit?

Ans — The statement is true.

Q2. Wi-Fi, Bluetooth, and cordless phones all use the same frequency ranges?

Ans — The statement is false.

Q3. Which wired network technology has an inner wire surrounded by a tubular insulating layer, that is then surrounded by a tubular conducting shield?

Ans — Coaxial cable.

6 Wireless Standards (802.11)

1. IEEE 802.11 governs the WLAN environment, collectively called "**Wi-Fi**".
2. The **Wi-Fi Alliance** tests wireless LAN devices from different manufacturers.

Router configuration settings:

- **Network Mode** → 802.11 b/g/n, or mixed mode.
 - **SSID** → the network name; all devices must use the same SSID to participate.
 - **Standard Channel** → default is Auto (AP automatically picks the best channel).
 - **SSID Broadcast** → default is Enabled (if disabled, the SSID must be entered manually).
4. All devices using the same standard = maximum speed achieved.
 5. **Mixed mode** allows devices with different Wi-Fi standards to connect together.
 6. **STA (Station)** = wireless devices; uses the SSID to identify which WLAN to join.

Q1. Which organization is responsible for testing wireless LAN devices?

Ans — Wi-Fi Alliance.

Q2. What is used to identify a specific wireless network?

Ans — The Service Set Identifier (SSID).

Q3. If you have devices on your wireless network using a variety of different 802.11 standards, should you set your network to the highest standard to get the best throughput?

Ans — The statement is false.

7 Setting Up a Home Router

1. **Wired setup:** Ethernet cable → from the computer's NIC to the router's LAN port.
2. **Link lights ON** = working connection.
3. The IP address is usually assigned automatically via **DHCP** from the router.
4. **SSID naming:** avoid using the device model/brand name — it's a security risk (internet searches can expose weaknesses).
5. **MAC address filtering** → allows only specific devices; makes the network more secure but less flexible when adding new devices.
6. **Guest access** → a separate SSID; allows internet-only access, isolated from the main network.

8 Module Based Important Questions — Module Test

Q1. Which type of wireless communication is based on 802.11 standards?

Ans — Wi-Fi.

Q2. What wireless router configuration would stop outsiders from using your home network?

Ans — Encryption.

Q3. What type of device is commonly connected to the Ethernet switch ports on a home wireless router?

Ans — LAN device.

Q4. Which type of network technology is used for low-speed communication between peripheral devices?

Ans — Bluetooth.

Q5. What can be used to allow visitor mobile devices to connect to a wireless network and restrict access of those devices to only the internet?

Ans — Guest SSID.

Q6. What purpose would a home user have for implementing Wi-Fi?

Ans — To create a wireless network usable by other devices.

Q7. What is another term for the internet port of a wireless router?

Ans — WAN port.

Q8. Which type of network cable consists of 4 pairs of twisted wires?

Ans — Category 5e.

Q9. What is the default SSID broadcast setting on a wireless router?

Ans — Enabled.

Q10. Which is a characteristic of the network SSID?

Ans — It is case sensitive.

CISCO NETWORKING BASICS

📄 CHECKPOINT EXAM — BUILD A SMALL NETWORK

📖 Checkpoint Exam

🔗 Covers Modules 1-4

⇒ Handwritten Notes → Digital

COURSE

Networking Basics

TYPE

Checkpoint Exam

STARTED

04 / 08 / 2026

FORMAT

20 Questions

0 About This Checkpoint

This checkpoint exam covers everything from **Modules 1-4** — network basics, data representation, bandwidth, end/intermediary devices, network media, ISP connections, wireless technology, and home router setup.

1 Checkpoint Exam — Questions & Answers

Q1. What data representation is used when a computer or network device is processing data?

Ans — Binary.

Q2. What allows digital devices to interconnect and transmit data?

Ans — A network.

Q3. How many unique values are possible using a single binary digit?

Ans — 2.

Q4. Which is a characteristic of the internet?

Ans — It is not centrally governed.

Q5. What is the order of bandwidth measurement from smallest to largest?

Ans — Kbps, Mbps, Gbps, Tbps.

Q6. Which two devices are considered end devices?

Ans — Laptop and printer are end devices.

Q7. Which items are collectively referred to as network media?

Ans — Wires and radio waves.

Q8. Define each device to a category — [PC, printer, firewall, smart device, router, switch]:

Ans — End devices: PC, printer, smart device. Intermediary devices: firewall, switch, router.

Q9. Which three devices are considered intermediary devices in a network?

Ans — Router, switch, wireless access point.

Q10. What are two types of wired high-speed internet connections?

Ans — Cable and DSL (Digital Subscriber Line).

Q11. What are two methods typically used on a mobile device to provide internet connectivity?

Ans — Cellular and Wi-Fi.

Q12. A user is looking for a wireless headphone for listening to songs stored on a smartphone. What wireless technology would the headphone use?

Ans — Bluetooth.

Q13. Which two methods are used to directly connect mobile devices such as tablets and smartphones to a data network?

Ans — Cellular communications and Wi-Fi.

Q14. In the context of mobile devices, what does the term tethering involve?

Ans — Connecting a mobile device to another mobile device or computer to share a network connection.

Q15. A consumer places a smartphone close to a pay terminal at a store and the shopping charge is successfully paid. Which type of wireless technology was used?

Ans — NFC (Near Field Communication).

Q16. A user is setting up a home wireless network. Which type of device must the user have in order to establish the wireless network and provide access to the internet for multiple home devices?

Ans — Wireless router.

Q17. Which technology is used to uniquely identify a WLAN network?

Ans — SSID (Service Set Identifier).

Q18. What two radio frequency bands are used in home wireless LANs?

Ans — 2.4 GHz and 5 GHz.

Q19. Which feature is characteristic of MAC filtering in wireless networks?

Ans — It restricts computer access to a wireless network.

Q20. Which wireless RF band do IEEE 802.11 b/g devices use?

Ans — 2.4 GHz.

CISCO NETWORKING BASICS

MODULE 5 — COMMUNICATION PRINCIPLES

Networking Fundamentals

Cisco Skills for All

Handwritten Notes → Digital

COURSE

Networking Basics

MODULE

5 of Course

STARTED

04 / 08 / 2026

FORMAT

Notes + Quiz + Test

0 Module Introduction

Module 5 explains the **rules that make network communication possible** — protocols, standards, and the layered models (TCP/IP and OSI) that describe how those rules are organized.

1 Communication Protocols — Fundamentals

Protocols are the rules required for computers to properly communicate across a network.

Protocols include:

- **Message format** — every message must use a specific format/structure.
- **Message size** — strict rules govern data-piece size, which can vary by channel.
- **Timing** — determines transmission speed, when a host can send, and how much data per transmission.
- **Encoding** — messages are converted into bits, then encoded as sound / light / electrical impulses depending on the media.
- **Encapsulation** — every message needs a header containing source + destination addressing info.
- **Message pattern** — some messages need acknowledgement (request/response); others simply stream without confirmation.

2 Practice Notes — Protocol Scenarios

Quick scenarios worked through while revising this topic:

Bianka & Nguyet — Scenario (order of protocols)

Method, language, confirmation.

Rory — MAC address order in Ethernet frame

Destination before source — protocol characteristic: encapsulation.

Rory — Ethernet frame size (64-1518 bytes → bits)

Two protocol characteristics: encoding, message size.

3 Communication Standards & Standards Organizations

1. **Topology** — a visual representation of end devices and intermediary devices in a network.
2. A device only "sees" its own addressing info — network protocols tell it whether another device is on the same network.
3. Communication is broken into smaller data units called **packets**.
4. A **standard** is a set of rules that determines how something must be done.
5. Standards ensure all devices implement the same protocol rules → different device types can communicate over the internet.
6. New standards go through a cycle of **discussion → problem-solving → testing**.
7. Each stage is recorded in a numbered **RFC (Request For Comments)** document.
8. RFCs are published and managed by the **IETF**.

Diagram — Network protocols:

Network Protocols

Ethernet

IP (Internet Protocol)

TCP

HTTP

Standards organizations that support the internet: IEEE, IETF, ITU, TIA, IANA (Internet Assigned Numbers Authority), ICANN (The Internet Corporation for Assigned Names and Numbers).

Rules governing network communications (format, size, timing, encapsulation) = network protocols. IETF records/publishes internet standards in documents known as Request For Comments (RFC).

4 Network Communication Models — TCP/IP

1. **Protocols** = rules that govern communications (e.g. HTTP, TCP, IP, Ethernet).
2. These protocols are implemented in software and hardware on every host/device.
3. **Protocol stack** = illustrates how protocols interact as a layered hierarchy. Each higher-level protocol depends on the services of the protocols below it. Layer separation enables each layer to operate independently.

TCP/IP model (4 layers) — used for actual internet communication:

Layer	Function
Application	Represents data to the user, plus encoding and dialog control.
Transport	Supports communication between devices across diverse networks.
Internet	Determines the best path through the network.
Network Access	Hardware devices and media that make up the network.

Reference model — describes the functions required at each layer, but not exactly how to accomplish them. It aids understanding, not implementation.

Protocol responsible for guaranteeing reliable data delivery:

TCP (Transmission Control Protocol).

Protocol used by routers to forward messages:

IP (Internet Protocol).

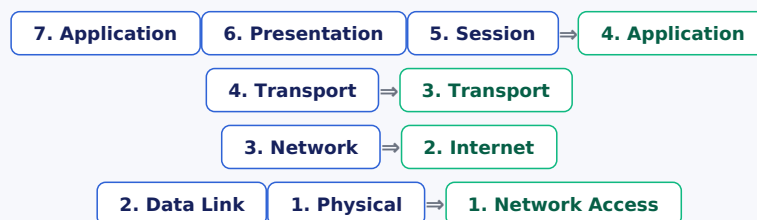
5 OSI 7-Layer Model

1. **OSI** = created by the OSI project at the international ISO.
2. Used for data network design, operation specifications, and troubleshooting.
3. **7 layers (top to bottom):**

#	Layer	Description
7	Application	Protocols for process-to-process communications.
6	Presentation	Common representation of data transferred between application layer services.
5	Session	Organizes dialogue and manages data exchange.
4	Transport	Segments, transfers, and reassembles data for individual communications between end devices.
3	Network	Exchanges individual pieces of data between identified end devices.
2	Data Link	Methods for exchanging data frames between devices over a common media.
1	Physical	Mechanical, electrical, functional, and procedural means to activate/maintain/deactivate physical connections.

6 OSI vs. TCP/IP Model — Comparison

OSI Model vs. TCP/IP Model — layer mapping:



OSI's top 3 layers → TCP/IP Application layer. OSI's bottom 2 layers → TCP/IP Network Access layer.

7 In-Chapter Practice Questions

Purpose of the OSI physical layer:

What is the purpose of the OSI physical layer?

Transmitting bits across the local media.

Correct statement about network protocols:

They define how messages are exchanged between the source and the destination.

Three acronyms representing standards organizations:

IANA, IETF, IEEE.

Term for rules at one layer governing communication at that layer:

Protocol.

OSI layer that segments/reassembles data for individual communications:

Transport.

Purpose of protocols in data communications:

Providing the rules required for a specific type of communication to occur.

8 Module Test — Questions & Answers

Q1. Term for rules defining how a network operates?

Ans — Protocol.

Q2. Three OSI layers that make up the TCP/IP application layer?

Ans — Presentation, application, session.

Q3. Organization that publishes/manages RFC documents?

Ans — IETF.

Q4. Two OSI layers equivalent to one TCP/IP layer?

Ans — Data link, physical (both map to Network Access).

Q5. OSI layer that segments/reassembles data?

Ans — Transport.

Q6. Protocol responsible for guaranteeing reliable data delivery?

Ans — TCP.

Q7. Protocol used by routers to forward messages?

Ans — IP.

Q8. Two OSI layers that map to the network access layer in the TCP/IP model?

Ans — Data link, physical.

Q9. IP addressing occurs at which layer of the OSI model?

Ans — Network.

Q10. Which organization created the OSI model?

Ans — The international ISO.

CISCO NETWORKING BASICS

MODULE 6 — NETWORK MEDIA

[← Networking Fundamentals](#)[🔗 Cisco Skills for All](#)[⇒ Handwritten Notes → Digital](#)

COURSE

Networking Basics

MODULE

6 of Course

STARTED

04 / 08 / 2026

FORMAT

Notes + Test

0 Module Introduction

Module 6 looks at **network media** — the physical (and wireless) channels that actually carry data between devices, the criteria used to choose the right media, and the three most common cable types used in networking.

1 Three Types of Network Media

1. **Metal wires within cables** → data encoded into electrical impulses.
2. **Glass/plastic fibers within cables** (fiber-optic) → data encoded into pulses of light.
3. **Wireless transmission** → data encoded via modulation of electromagnetic wave frequencies.

2 Four Criteria for Choosing Media

1. Maximum **distance** the media can successfully carry a signal.
2. **Environment** in which the media will be installed.
3. Amount of **data and speed** at which it must be transmitted.
4. **Cost** of the media installation.

3 Three Most Common Network Cables

1. **Twisted-pair cable** → used by Ethernet to interconnect devices.
2. **Coaxial cable** → copper cable used by cable TV companies and satellite communication systems.
3. **Fiber-optic cable** → glass/plastic, thin as human hair, carries data at very high speed over long distances; unaffected by electrical interference since it uses light.

4 Quick Notes — Key Facts

Not a criterion for choosing network media:

The type of data that can be transmitted.

Most common wired media used by TV companies (set-top box):

Coaxial cable.

Wired media for very high-speed digital info over long distance:

Fiber-optic cable.

Most common wired media to connect end devices to an Ethernet LAN:

Twisted-pair cable.

5 Network Media in 5 Points (Recap)

1. **Media types** — networks use 3 main media: metal wires (electrical impulses), fiber-optic cables (light pulses), and wireless transmission (electromagnetic wave modulation).
2. **Media definition** — communication transmits across a network via media, which provides the channel for messages to travel from source to destination.

3. **Selection criteria** — media choice depends on maximum signal distance, installation environment, data volume/speed required, and installation cost.
4. **Cable types** — twisted-pair (used by Ethernet), coaxial cable (used by cable TV/satellite systems), and fiber-optic (glass/plastic, hair-thin, high speed, immune to electrical interference since it uses light).
5. **Key advantage** — fiber-optic is the fastest and most interference-resistant option for long-distance, high-speed data transmission compared to copper-based media.

6 Module 06 — Questions & Answers

Q1. Cable for high-speed video surveillance over hundreds of meters?

Ans — Fiber-optics.

Q2. Two common network media?

Ans — Fiber, copper.

Q3. Cable commonly used to connect office computers to a local network?

Ans — Twisted-pair cable.

Q4. Three factors for choosing network media?

Ans — Distance between hosts, amount of data/transfer rate, environment where media is installed.

Q5. Why is fiber preferable to copper (3 reasons)?

Ans — Greater bandwidth potential, greater distance per cable run, limited susceptibility to EMI/RFI.

Q6. Media carrying data as electrical impulses?

Ans — Copper cable.

Q7. Two media types carrying data as electrical impulses?

Ans — Twisted-pair cable, coaxial cable.

Q8. Media carrying data as impulses of light?

Ans — Fiber-optic cable.

Q9. Best media for a small office local network upgrade?

Ans — Twisted-pair cable.

Q10. Purpose of twisted pairs in Ethernet cable?

Ans — To reduce interference.

CISCO NETWORKING BASICS

MODULE 7 — THE ACCESS LAYER

Networking Fundamentals Cisco Skills for All Handwritten Notes → Digital

COURSE	MODULE	STARTED	FORMAT
Networking Basics	7 of Course	10 / 08 / 2026	Notes + Quiz + Test

0 Module Introduction

Module 7 zooms into the **access layer** — where hosts actually plug into the network. It covers the structure of an Ethernet frame, how switches build and use MAC address tables, and why switches replaced hubs.

1 Ethernet Frame & Encapsulation

Ethernet frames travel over a LAN using an **Ethernet NIC** (Network Interface Card).

Ethernet frame structure:

Field	Size
Preamble	7 bytes
Start Frame Delimiter	1 byte
Destination MAC address	6 bytes
Source MAC address	6 bytes
Length/Type	2 bytes
Data	46-1500 bytes
Frame Check Sequence (FCS)	4 bytes

Encapsulation is the process of placing one message format inside another (like a letter inside an envelope) so it can be delivered over a network. **De-encapsulation** reverses this at the receiving end.

Ethernet protocol standards define frame format, frame size, timing, and encoding for network communication. An IPv6 packet header includes fields like Version, Traffic Class, Flow Label, Payload Length, Next Header, Hop Limit, Source IP address, and Destination IP address (40 bytes total).

2 Hubs vs. Switches

The access layer is where hosts connect to the network to access other hosts, shared files, and printers — via an Ethernet cable to an access layer device.

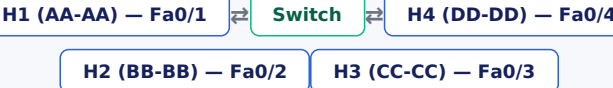
Ethernet hubs are obsolete — they let only one message pass at a time, and simultaneous messages cause collisions and network slowdowns.

Ethernet switches replaced hubs — they work at Layer 2, read the MAC address from frames, and build a MAC address table to connect source and destination through a temporary circuit.

3 MAC Address Table — Worked Example

Port	MAC Address
Fa0/1	AA-AA
Fa0/2	BB-BB
Fa0/3	CC-CC
Fa0/4	DD-DD

Note: "Fa" stands for Fast Ethernet.



Switch forwards using its MAC address table, mapping each port to a host's MAC address.

Switches support simultaneous send/receive on the same cable (eliminating collisions), and the MAC address table updates dynamically as new source MAC addresses are read.

4 In-Chapter Practice Questions

Q1. Ethernet protocol operates at which layer of the OSI model?

Ans — Layer 2 (Data Link).

Q2. Are Ethernet hubs considered obsolete?

Ans — Yes.

Q3. What do switches base their forwarding decision on?

Ans — The destination MAC address field.

Q4. What indicate the beginning of an Ethernet frame?

Ans — Preamble and SFD (Start Frame Delimiter).

Q5. If the destination MAC address is not in the switch's table, what does the switch do?

Ans — It forwards the frame out all ports except the incoming port.

Q6. What is prepending protocol info with info from another protocol called?

Ans — Encapsulation.

Q7. How do switches add new entries to the MAC address table?

Ans — Based on the source MAC address field.

Q8. What does the destination MAC address indicate?

Ans — The MAC address of the device on the network that will receive the frame.

5 Module Test — Questions & Answers

Q1. When the destination MAC address is not in the MAC table, what does a Layer 2 switch do?

Ans — Forwards the frame out all ports except the one it was received on.

Q2. What network device sends data to a specific destination based on its MAC address table?

Ans — A switch.

Q3. How does a switch build its MAC address table?

Ans — Using the source Layer 2 (MAC) address of incoming frames.

Q4. What is the FCS field used for?

Ans — To determine if errors occurred during frame transmission and reception (via CRC comparison).

Q5. What is one function of a Layer 2 switch?

Ans — To determine which interface to forward a frame on, based on the destination MAC address.

Q6. How does a switch keep its MAC address table current?

Ans — Using the source MAC address and the incoming port.

Q7. What is the process of placing one message inside another for transfer called?

Ans — Encapsulation.

Q8. What does an 802.3 Ethernet frame include?

Ans — Destination physical (MAC) address, frame check sequence (FCS), and source physical (MAC) address.

Q9. If a host receives a unicast frame with a destination MAC that doesn't match its own, what does it do?

Ans — It discards the frame.

Q10. What are Ethernet switch frame-forwarding decisions based on?

Ans — MAC address and port mappings in the MAC address table.

Q11. Exhibit (PC1-PC3): Since PC3's MAC address is not in the table, what will the switch do?

Ans — Forward the frame to all ports except the port it was received on.

CISCO NETWORKING BASICS

📄 CHECKPOINT EXAM — NETWORK ACCESS

📖 Checkpoint Exam

🔄 Covers Module 7

⇒ Handwritten Notes → Digital

COURSE

Networking Basics

TYPE

Checkpoint Exam

STARTED

11 / 08 / 2026

FORMAT

Bullet-Point Notes

0 About This Checkpoint

This checkpoint exam covers **Module 7 — The Access Layer**: Ethernet protocols and models, communication standards, network media, encapsulation and switching behavior.

1 Protocols & Models — Recap

1. Scenario (order of protocols) — Bianka & Nguyet: method, language, confirmation.
2. MAC address order in Ethernet frame (destination before source) — protocol characteristic: encapsulation.
3. Ethernet frame size (64-1518 bytes → bits) = two protocol characteristics: encoding, message size.
4. Application protocol = governs web server-client interaction; Internet protocol = encapsulates segments into packets + addressing; Network access protocol = prepares packets for media transmission; Transport protocol = manages individual server-client conversations.
5. Standards organizations = IEEE, IETF, IANA.
6. All communication methods share: source, destination, transmission medium.
7. TCP/IP layers — Network access: controls hardware/media. Internet: best path routing. Application: represents data/encoding/dialog. Transport: supports communication between devices.

2 Network Media — Recap

1. Criterion for choosing media = installation environment.
2. Fiber-optic = data encoded as pulses of light.
3. Coaxial cable uses = satellite systems + home TV connections.
4. Fiber optic real-world considerations = needs different termination/splicing skills + higher capacity but costlier than copper (not RFI/EMI affected — light based, so immune).
5. Cable for backbone/telephone = fiber-optic.
6. Fiber's advantage over copper = carries signal much farther.

3 Encapsulation & Switching — Recap

1. Placing one message format inside another = encapsulation.
2. Empty MAC table (SW1 → PC-C) = SW1 floods the frame on all its ports, except the one it entered on (never dropped).
3. New MAC address on a port = switch adds the source MAC to the MAC table first.
4. A Cisco switch's 2 core actions = builds the MAC table using the source MAC + forwards frames using the destination MAC via that table.
5. Switch builds MAC table using = source MAC address.
6. Normal Ethernet frame carries = 46 to 1500 bytes of data.
7. Switch populates the MAC table using = source MAC address + incoming port.

4 Quick Memory Hooks

Source MAC = table banane ke liye; Destination MAC = to frame forward karne ke liye using...

The forwarding decision.

Switch has never drop frame when destination is unknown —

Always work floods instead of except incoming port.

Fiber optics cable is light based and EMI/RFI proof, but...

Skill and cost are expensive.

CISCO NETWORKING BASICS

MODULE 8 — THE INTERNET PROTOCOL

🔗 Networking Fundamentals 🔗 Cisco Skills for All ➡ Handwritten Notes → Digital

COURSE Networking Basics	MODULE 8 of Course	STARTED 12 / 08 / 2026	FORMAT Notes + Packet Tracer + Test
-----------------------------	-----------------------	---------------------------	--

0 Module Introduction

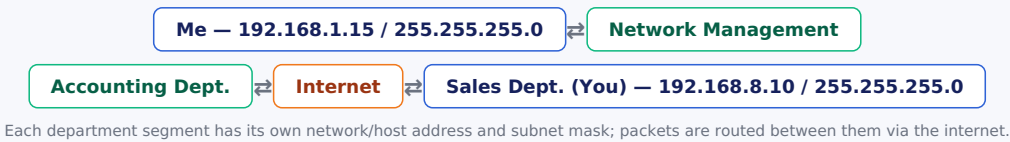
Module 8 covers the **Internet Protocol (IP)** — how packets actually travel across a network using NIC, IP addressing, and TTL, and walks through a Packet Tracer simulation of connecting to a web server, then introduces the basics of IPv4 addressing and subnet masks.

1 The Internet Protocol & Packet Travel

The internet protocol uses the NIC.

NIC = Network Interface Card. **TTL** = Time to Live.

Diagram — Packet travel & tracer through the internet:



2 Packet Tracer — Connect to a Web Server

1. Packet Tracer is Cisco's network simulation software used to build and test virtual networks.
2. Setup: PC0 (client) connects to a web server with IP address 172.33.100.50.
3. Ping command (ping 172.33.100.50) checks connectivity between client and server.
4. ARP (Address Resolution Protocol) resolves the server's MAC address before the first packet is sent.
5. ICMP (Internet Control Message Protocol) Echo Request and Echo Reply are the messages exchanged during a ping.
6. Ping result: 4 packets sent, 3 received, 1 lost (25% loss) — first packet lost due to ARP delay.
7. TTL (Time to Live) limits how many hops (routers) a packet can pass through before expiring.
8. Web browser sends an HTTP request to fetch webpage data using the server's IP address.
9. Server responds with an HTTP response containing the webpage's HTML content.
10. Final result: webpage loads with the message "Welcome to the Learn IP website."
11. Simulation Mode in Packet Tracer visually shows packet travel between devices, layer by layer.
12. These concepts (ARP, ping, TTL, HTTP flow) are directly useful for SOC Analyst work — especially reading Wireshark logs and analyzing network traffic.

3 IPv4 Addressing — First Look

Same address, different subnet mask — same or different network?

Host A	Comparison Address	Same Network?
10.5.4.100 / 255.255.255.0	Network address = 10.5.4.0	10.5.4.99 → Yes 10.5.4.1 → Yes
172.16.4.100 / 255.255.0.0	Network address = 172.16.0.0	172.16.0.1, 172.16.4.99 → Yes
192.168.1.50 / 255.255.255.0	Network address = 192.168.1.0	192.168.1.100, 192.168.1.1 → Yes

Purpose of the IPv4 address:

1. An IPv4 address is a logical network address that identifies a particular host.
2. Must be properly configured and unique within the LAN for local communication.
3. Must also be properly configured and unique in the world for remote (internet) communication.
4. An IPv4 address is assigned to the network interface connection of a host — usually a NIC installed in the device.
5. Every packet sent across the internet has a source and destination IPv4 address.
6. This info is required by networking devices to deliver the packet to the destination and return replies to the source.

The IPv4 Address Structure:

1. The IPv4 address is a logical 32-bit address, and it is hierarchical.
2. Made up of two parts: Network portion, Host portion.

Example: IP = 192.168.5.11, Subnet mask = 255.255.255.0

- First 3 octets (192.168.5) = Network portion.
 - Last octet (11) = Host portion.
4. This is called hierarchical addressing — the network portion tells which network the host address belongs to.
 5. Routers only need to know how to reach each network, not the location of every individual host.
 6. With IPv4 addressing, multiple logical networks can exist on one physical network, as long as the network portion differs.

5 Module Recap — Bullet Points

1. Design criterion for IPv4 addressing → each IP address must be unique within the local network.
2. 4 octets make up an IPv4 address.
3. Network portion and host portion are components of an IPv4 address.
4. To determine the subnet a host belongs to → purpose of subnet mask.
5. IP phone, printer with integrated NIC, and server with two NICs are devices needing IP addresses.
6. A physical network can connect multiple devices of different IPv4 logical networks → defines the physical vs. logical IPv4 network relationship.
7. *Why?* One physical network can host multiple logical (IP) networks.
8. 32 bits is the size of an IPv4 address.
9. Network number of 172.16.34.10 / 255.255.255.0 → 172.16.34.0.
10. An IPv4 addressing scheme is hierarchical, and IPv4 is a logical addressing scheme — two features of IPv4 addresses.
11. *Why?* IPv4 addresses work on both local networks, not internet only.
12. Same local network (255.255.255.0) → 192.168.10.56, 192.168.10.2.
13. Routing technology connects different logical networks.
14. *Why?* Routing is required for inter-network (different logical network) communication.

CISCO NETWORKING BASICS

MODULE 9 — IPv4 AND NETWORK SEGMENTATION

Networking Fundamentals Cisco Skills for All Handwritten Notes → Digital

COURSE	MODULE	STARTED	FORMAT
Networking Basics	9 of Course	13 / 08 / 2026	Notes + Quiz + Test

0 Module Introduction

Module 9 is a deep dive into **IPv4 addressing** — how to recognize private, public, loopback, link-local, multicast, and experimental addresses; the three transmission types (unicast, broadcast, multicast); classful addressing history; and why large networks are broken into smaller **subnets** through network segmentation.

1 Recognizing IPv4 Address Types

Private IP ranges — used only inside local networks, never routed directly on the internet:

Range	Notes
10.0.0.0 – 10.255.255.255	Class A private block (/8)
172.16.0.0 – 172.31.255.255	2nd octet must be 16–31
192.168.0.0 – 192.168.255.255	Class C private block (/16)

Public IP — directly reachable on the internet; any address that does not fall into private or the special ranges below.

Loopback IP — a device testing itself ("localhost"): 127.0.0.0 – 127.255.255.255 (most common: 127.0.0.1).

Local-Link / APIPA — auto-assigned by the OS when DHCP fails: 169.254.0.0 – 169.254.255.255.

DHCP = Dynamic Host Configuration Protocol. APIPA = Automatic Private Internet Protocol Addressing.

Multicast — one sender → a specific group of receivers: 224.0.0.0 – 239.255.255.255.

Experimental — reserved for research/future use only: 240.0.0.0 – 255.255.255.255.

Recognition Rule (check in this exact order):

- Is the first octet 127? → Loopback.
- Are the first two octets 169.254? → Link-Local.
- Is the first octet 224–239? → Multicast.
- Is the first octet 240–255? → Experimental.
- Does it fall in one of the 3 private ranges? → Private.
- None of the above → Public.

Quick-glance table (first octet only):

First Octet	Condition	Type
10	Always	Private
127	Always	Loopback
169	Only if .254 follows	Link-Local
172	2nd octet 16–31	Private
192	2nd octet fixed (168)	Private
224–239	—	Multicast
240–255	—	Experimental
Anything else	—	Public

Worked examples:

- 192.168.1.5 → Private (matches 192.168.x.x exactly).
- 192.169.1.1 → Public (not 192.168 — one digit changes everything).
- 8.8.8.8 → Public (Google DNS — no special range matches). DNS = Domain Name System.
- 10.200.5.3 → Private (first octet is 10).
- 172.20.5.1 → Private (2nd octet 20 is within 16–31).
- 172.45.1.1 → Public (2nd octet 45 is outside 16–31).
- 127.0.0.1 → Loopback.
- 169.254.10.20 → Link-Local.
- 224.0.0.5 → Multicast.

10. 240.1.1.1 → Experimental.

2 Transmission Types — Unicast, Broadcast, Multicast

Transmission types (a separate concept from IP category):

(I) Unicast — one sender → one receiver (like a personal phone call).

Source: 172.16.4.1/24 → Switch → Destination: 172.16.4.253/24

Only the addressed host (172.16.4.253) processes the packet — the rest ignore it.

(II) Broadcast — one device sends a message to all devices on a network (one-to-all communication). Uses the subnet's last address, or 255.255.255.255.

Source: 172.16.4.1/24 → 255.255.255.255 (Limited Broadcast)

All hosts in the broadcast domain receive and process the frame.

(III) Multicast — one sender → a specific group (like a WhatsApp group message). Range: 224.0.0.0 – 239.255.255.255.

Source: 172.16.4.1/24 → 224.10.10.5 (Multicast Group)

Only clients subscribed to the multicast group accept the packet. OSPF = Open Shortest Path First.

3 Common Confusion Points

1. 192.168.x.x = private, but 192.169.x.x = public — only exactly .168 is private.
2. 224–239 = Multicast, 240–255 = Experimental — adjacent ranges, easy to mix up.
3. A "big" or unfamiliar-looking number does NOT automatically mean public or private — always check the actual range, never guess by feel.

4 IPv4 Address Exhaustion & Transmission Types (Detail)

1. Private IPv4 addresses can be used by any organization (home, school, office, company) via 10.0.0.0/8, and are assigned within an organization's intranet (internal network).
2. Public IPv4 address exhaustion is why private IPv4 addresses exist, and organizations are transitioning to IPv6; also, accessing a device over the internet requires the destination IPv4 address to be public.
3. **RIR** (Regional Internet Registry) is the organization that receives IP addresses from IANA and allocates them to ISPs and some organizations.

IANA = Internet Assigned Numbers Authority.

IPv4 Unicast, Broadcast and Multicast:

(A) Unicast — one device sends a message to one other device (one-to-one communication).

- A unicast packet's destination IP is a single recipient's address.
- Source IP is always unicast (a packet can only originate from one source), regardless of whether the destination is unicast, broadcast, or multicast.
- IPv4 unicast host address range: 1.1.1.1 to 223.255.255.255.

(B) Broadcast — one device sends a message to all devices on a network (one-to-all communication).

- A broadcast packet has all 1s (32 one-bits) in the host portion of the destination address.
- Broadcast packets must be processed by every device in the same broadcast domain.
- Two types: **Directed broadcast** — sent to all on a specific network. **Limited broadcast** — sent to 255.255.255.255.
- By default, routers do NOT forward broadcasts.

(C) Multicast — one device sends a single packet to a selected group of hosts subscribed to a multicast group (reduces network traffic).

- Multicast address range: 224.0.0.0 to 239.255.255.255.
- Each multicast group has one shared IPv4 multicast destination address.
- A host subscribed to a multicast group processes packets addressed to both the multicast address AND its own unique unicast address.

5 Types of IPv4 Addresses & Classful Addressing

Types of IPv4 addresses:

(A) Public IPv4 addresses — globally routed between ISP routers, usable directly on the internet, must be unique.

- (B) Private IPv4 addresses** — used by most organizations for internal hosts (intranet), NOT globally routable. Before reaching the internet, a private address must be translated to a public address using **NAT** (Network Address Translation).
- (C) Loopback addresses** — 127.0.0.0/8 (range 127.0.0.0 – 127.255.255.254), commonly identified as just 127.0.0.1 — used by a host to direct traffic to itself.
- (D) Link-Local addresses** — 169.254.0.0/16 (range 169.254.0.1 – 169.254.255.255), known as APIPA — self-assigned by a Windows DHCP client when no DHCP server is available.

(E) Classful Addressing (RFC 790, 1981):

Class	Range	Description
A	0.0.0.0/8 – 127.0.0.0/8	Extremely large networks (16+ million host addresses)
B	128.0.0.0/16 – 191.255.0.0/16	Moderate to large networks (~65,000 host addresses)
C	192.0.0.0/24 – 223.255.255.0/24	Small networks (max 254 hosts)
D (multicast)	224.0.0.0 – 239.0.0.0	Multicast block
E (experimental)	240.0.0.0 – 255.0.0.0	Experimental block

Both IPv4 and IPv6 addresses are managed by IANA. IANA allocates IP address blocks to RIRs (Regional Internet Registries), who allocate addresses to ISPs (who provide address blocks to organizations and smaller ISPs). Organizations can also obtain addresses directly from an RIR.

Five RIR organizations:

RIR	Full Form	Region
ARIN	American Registry for Internet Numbers	North America
AFRINIC	African Network Information Center	Africa
APNIC	Asia-Pacific Network Information Center	Asia/Pacific
LACNIC	Latin American and Caribbean IP Address Registry	Latin America & Caribbean
RIPE NCC	Réseaux IP Européens Network Coordination Center	Europe, Middle East, Central Asia

Classful address distribution (approx.): Class A – 50%, Class B – 25%, Class C – 12.5%, Class D + E – 12.5%.

Class	Total Networks	Total Hosts / Net
A	128	16,777,214
B	16,384	65,534
C	2,097,152	254

6 Broadcast Containment & Network Segmentation

1. Routers do not forward an IPv4 broadcast packet by default (Ethernet switches and Windows PCs don't have this default blocking behavior).
2. Excessive broadcast traffic results in slow network operations and slow device operations.

Network Segmentation:

4. In an Ethernet LAN, devices use broadcasts and ARP to locate other devices.
5. ARP sends a Layer-2 broadcast to a known IPv4 address to discover the associated MAC address. *ARP = Address Resolution Protocol. MAC = Media Access Control.*
6. A host typically acquires its IPv4 configuration using DHCP, which sends broadcasts on the local network to locate a DHCP server.
7. Switches propagate a broadcast out all interfaces except the one on which it was received.
8. A large broadcast domain connects many hosts — problem: excessive broadcasts can negatively affect network performance.
9. Solution: reduce network size into smaller broadcast domains — a process called **subnetting**.
10. Smaller resulting network spaces are called subnets.
11. Basis of subnetting: using host bits to create additional subnets.

Benefits of subnetting / network segmentation:

- Reduces overall network traffic.
- Improves network performance.
- Helps administrators implement security policies (which subnets can/cannot communicate).
- Reduces the number of devices affected by abnormal broadcast traffic from misconfigurations, hardware/software problems, or malicious intent.

7 Module Test — Key Points Recap

1. Subnet mask → identifies which network a host belongs to.
2. Subnetting → reduces broadcast flooding scope.
3. Message to all hosts on a remote network → directed broadcast.
4. IP 169.254.0.0 → Link-Local address.

5. Private IPs → 10.0.0.0, 172.16–31.0.0, 192.168.0.0.
6. 240.2.6.255 → Experimental; 169.254.1.5 → Link-Local; 198.133.219.2 → Public; 127.0.0.1 → Loopback.
7. Router → divides Layer-2 broadcast domains.
8. IANA → manages allocation of IP addresses & domain names.
9. IPv4 multicast range → 224.0.0.0 – 239.255.255.255.
10. 192.168.25.10 unreachable → it's a private IP, needs NAT.
11. Broadcast → received by all hosts in the same network.

NAT = Network Address Translation.

CISCO NETWORKING BASICS

MODULE 10 — IPv6 ADDRESSING FORMATS AND RULES

Networking Fundamentals Cisco Skills for All Handwritten Notes → Digital

COURSE	MODULE	STARTED	FORMAT
Networking Basics	10 of Course	14 / 08 / 2026	Notes + Quiz + Test

0 Module Introduction

Module 10 covers **IPv6** — why it was created, how its 128-bit addresses are written and shortened, the different address types, and how IPv4 and IPv6 will coexist during the migration period.

1 Why IPv6? Coexistence & Notation Rules

- IPv4 depletion is the main reason for moving to IPv6, which has a 128-bit address space (340 undecillion addresses).
- ICMPv6 improves on ICMPv4 by adding address resolution and autoconfiguration.
- IPv4 and IPv6 will coexist for years; migration uses 3 techniques: **Dual Stack** (both protocols run together), **Tunneling** (IPv6 packet encapsulated inside IPv4), and **Translation / NAT64** (converts IPv6 ↔ IPv4 packets).
- IPv6 addresses are 128 bits, written as 32 hexadecimal digits in 8 hextets (each hextet = 16 bits), and are not case-sensitive.
- Preferred format = writing all 32 hex digits fully (e.g., fe80:0000:0000:0123:4567:89ab:cdef).
- Rule 1 (omit leading zeros):** leading zeros in a hextet can be dropped (e.g., 09F0 → 9F0), but trailing zeros cannot be omitted.
- Rule 2 (double colon ::):** a :: can replace one contiguous string of all-zero hextets, but can be used only once in an address; if multiple such strings exist, use :: on the longest one (or the first, if equal length).

Note: Most important motivating factor for moving to IPv6 → depletion of IPv4 addresses. 4 out of 5 RIRs no longer have enough IPv4 to allocate regularly. Dual Stack technique using native IPv6 connectivity.

RIR = Regional Internet Registry | **IETF** = Internet Engineering Task Force | **ICMP** = Internet Control Message Protocol | **NAT** = Network Address Translation | **ISP** = Internet Service Provider | **IoT** = Internet of Things

2 IPv4/IPv6 Coexistence Diagrams

Diagram — IPv4 and IPv6 coexistence:

- Fig 1 — Dual Stack:** PC1, PC2, PC3 all connect through a switch running both IPv4 and IPv6 stacks simultaneously.
- Fig 2 — Tunneling:** PC1 (IPv6-only network) — S1 (Dual Stack) — [Tunnel through IPv4-only network] — S2 (Dual Stack) — PC2 (IPv6-only network).
- Fig 3 — Translation:** PC1 (IPv6-only network) ↔ NAT64 Router ↔ PC2 (IPv4-only network).

Bit segments (hextets) of 16 bits each:



3 Types of IPv6 Addresses

Types of IPv6 addresses (categories):

Type	Description	Range
Unicast	One sender, one receiver (specific).	—
Global Unicast (GUA)	Public, directly routable on the internet (similar to IPv4's "public IP").	Starts from 2000::/3
Link-Local	Works only within the local network segment; never forwarded by a router.	FE80::/10
Unique Local (ULA)	Similar to IPv4's "private IP" concept; used only within an organization, not routed on the internet.	FC00::/7 (in practice, mostly appears as FD00::/8)
Loopback	Used by a device to test itself (equivalent to IPv4's 127.0.0.1).	::1 (the only loopback address that exists)

Multicast	One-to-many communication; data sent to a specific group of devices.	FF00::/8
Anycast	A new concept introduced in IPv6 (not commonly present in IPv4). Multiple devices share the same address; the packet is delivered to the nearest device (commonly used in CDN servers and DNS root servers).	No separate range — it's a unicast address used in a special configuration.

Important: IPv6 does not have a broadcast type. Multicast is used instead to achieve one-to-many communication.

4 RIR Exhaustion Dates & Recognition Trick

The five global RIRs and their IPv4 exhaustion dates:

RIR	Region	IPv4 Exhaustion Date
APNIC	Asia-Pacific	June 2014
ARIN	North America	July 2015
RIPE NCC	Europe, Middle East, Central Asia	September 2012
LACNIC	Latin America & Caribbean	April 2011
AFRINIC	Africa	2020

Instant recognition trick — just look at the starting hexadecimal group to identify the address type:

Starts With	Type
::1 (entire address)	Loopback
FE80	Link-Local
FC00 or FD00	Unique Local (private-like)
FF (i.e. FF00–FFFF)	Multicast
2 or 3 (i.e. 2000–3FFF)	Global Unicast (public)

Quick example practice:

- FE80::1 → Link-Local
- 2001:db8::5 → Global Unicast (public)
- FF02::1 → Multicast
- ::1 → Loopback
- FD12:3456::1 → Unique Local (private)

5 Module Test — Key Points Recap

1. Advantage of IPv6 → more addresses for networks and hosts.
2. Reason for IPv6 creation → to relieve IPv4 address depletion.
3. Hex value of decimal 15 is F.
4. Device with both IPv4 & IPv6 on same adapter → Dual Stack.
5. Two ways to shorten IPv6 notation → omit leading zeros; use :: for all-zero hextets.
6. Encapsulates IPv6 inside IPv4 packets → tunneling.
7. Double colon (::) represents → continuous string of one or more all-zero hextets.
8. IPv6 address size is 128 bits.
9. Enables IPv6-only ↔ IPv4-only communication → NAT64.
10. Valid IPv6 notation → 2001:DB8:0:1111::200.
11. Correct about IPv4/IPv6 → IPv6 uses hex numbers; IPv4 is 32 bits.

CISCO NETWORKING BASICS

MODULE 11 — DYNAMIC ADDRESSING WITH DHCP

🔗 Networking Fundamentals 🔗 Cisco Skills for All ➡ Handwritten Notes → Digital

COURSE	MODULE	STARTED	FORMAT
Networking Basics	11 of Course	15 / 08 / 2026	Notes + Quiz + Test

0 Module Introduction

Module 11 covers **DHCP** (Dynamic Host Configuration Protocol) — why manual (static) addressing doesn't scale, the DORA process that automatically assigns IP addresses, DHCP leases, and how home routers configure DHCP by default.

1 What is DHCP & Why It's Necessary

What is DHCP? A protocol that automatically assigns IP addresses and network configuration to devices connecting to a network.

Without DHCP, every device would need IP, subnet mask, gateway, and DNS configured manually (static) — impractical on a large network.

Why DHCP is necessary:

- Manually assigning IPs is time-consuming and error-prone (risk of duplicate IPs).
- DHCP automatically manages which IP is given to which device, preventing conflicts.
- When a device disconnects, its IP is automatically freed for reuse by another device.

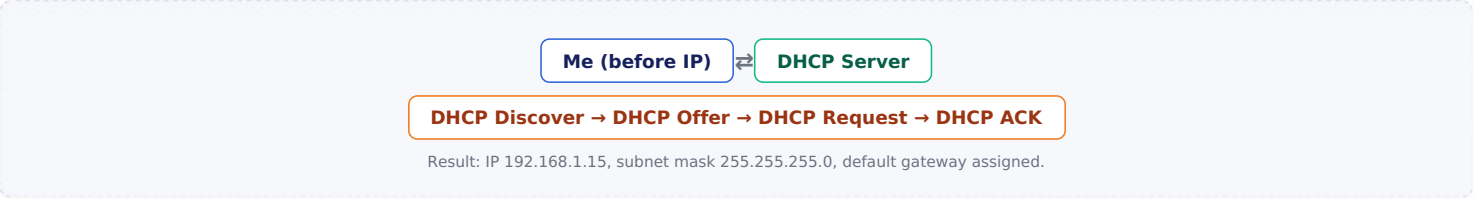
2 The DORA Process

DORA process (4 steps of DHCP):

Step	Description
Discover	Client broadcasts "I need an IP, is there a DHCP Server?"
Offer	Server offers a free IP from its available pool.
Request	Client accepts the offer and requests confirmation.
Acknowledge	Server confirms; client starts using the IP.

Memory trick: D → O → R → A (Discover → Offer → Request → Acknowledge).

Diagram — how the DHCP process works:



3 DHCP Lease & Server Location

DHCP lease:

- An IP address is assigned for a fixed lease period (e.g. 24 hours, 7 days — set by admin), not permanently.
- The client automatically sends a renewal request before the lease expires.
- If a device disconnects and its lease expires, the IP can be reassigned to another device.

Where the DHCP server is located:

- **Home network:** the router usually acts as the DHCP server.
- **Office/enterprise network:** a dedicated DHCP server is used (e.g. Windows Server or a Linux-based server).

DHCP Relay (advanced concept): used when the DHCP server is on a different subnet/broadcast domain than the client. A DHCP Relay agent forwards the client's broadcast request to the DHCP server on another subnet.

Connection to APIPA: if DHCP is unavailable or fails, a Windows device self-assigns an APIPA address (169.254.x.x). This allows basic local communication only — no internet access.

APIPA = Automatic Private Internet Protocol Addressing.

4 Static vs. Dynamic Addressing — Deep Dive

Static addressing:

- Network administrator manually configures each host (IPv4 address, subnet mask, default gateway).
- Gives more control, but is time-consuming; must maintain an accurate list of assigned addresses.

Dynamic addressing (DHCP):

- Protocol that automatically assigns IPv4 addresses.
- Preferred for large networks — reduces staff burden, eliminates entry errors.
- Address is leased, not permanent — returned to the pool when the host powers down/leaves.

Example: wireless hotspot → laptop's DHCP client contacts the local DHCP server → gets an IPv4 address.

Home networks: the DHCP server may be at the ISP, or the wireless router itself acts as both DHCP client and server.

DHCP process (4 steps — DORA), with technical detail:

- **Discover:** client broadcasts DHCP Discover (destination IP: 255.255.255.255, destination MAC: ff-ff-ff-ff-ff-ff).
- **Offer:** DHCP server replies suggesting an IPv4 address.
- **Request:** client sends DHCP Request to use that address.
- **Acknowledgment:** server sends DHCP Acknowledgement (confirms lease).

Home router DHCP config:

- Access via browser using the router's default IPv4 (usually 192.168.0.1).
- Default subnet mask: 255.255.255.0.
- Router IP = default gateway + internal DHCP server address.
- DHCP server is enabled by default on most home wireless routers.

5 Module Test — Questions & Answers

Client finds server

Called DHCP Discover; server responds first — called DHCP Offer; through DHCP Request the client accepts the offered IP; then DHCP ACK is called server confirms lease.

Reduce burden on staff and eliminates address config errors

DHCP preferred for large networks.

Reply to DHCP offer

DHCP request.

DHCP discover destination IP

255.255.255.255.

Server's reply after discover

DHCP offer.

DHCP advantage for mobile devices

Address leases temporary and returned to pool when device is off.

DHCP pool 192.168.0.100-149 — first device gets?

192.168.0.100.

PC1 rebooting after 2 weeks — destination IP?

255.255.255.255.

Server type that dynamically assigns IP

DHCP.

Message confirming client can use IP

DHCP Ack.

DHCP discover true statements

Comes from a client seeking an IP; all hosts receive it but only the DHCP server replies; destination IP is 255.255.255.255.

CISCO NETWORKING BASICS

📄 CHECKPOINT EXAM — THE INTERNET PROTOCOL

📖 Checkpoint Exam

🔄 Covers Modules 8-11

➡ Handwritten Notes → Digital

COURSE
Networking Basics

TYPE
Checkpoint Exam

STARTED
16 / 08 / 2026

FORMAT
20 Questions

0 About This Checkpoint

This checkpoint exam covers **Modules 8-11** — IP addressing basics, subnetting and private addressing, IPv6 addressing, and DHCP.

1 IP Addressing Basics

IP Addressing Basics:

Host 10.10.10.2 with subnet mask 255.255.255.0 — host portion?

Ans — 2.

Subnet mask identifies

Ans — The part of the IP address that identifies the network.

Network number of shown LAN

Ans — 192.168.10.0

Same logical network (mask /24)

Ans — 172.16.45.35, 172.16.45.12, and 172.16.45.56.

Bits in an IPv4 address

Ans — 32 bits.

Purpose of subnet mask and IP

Ans — Determines the subnet a host belongs to.

2 Subnetting & Private Addressing

Two reasons to create subnets

Ans — Easier to implement security policies, and improves network performance.

Three private IP ranges

Ans — 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.

127.0.0.1 / 169.254.1.5 / 172.19.20.5 / 240.2.6.255

Ans — Loopback; Link-Local; Private; Experimental.

PC got IP 169.254.100.88 — meaning?

Ans — Unreachable DHCP server; Windows falls back to the APIPA range (169.254.0.0 - 169.254.255.255).

3 IPv6 Addressing

Shortened form of 2001:0db8:0000:0000:ab00:0000:0000:0000

Ans — 2001:db8:0:0:ab00::

Most compressed form of 2001:0DB8:0000:AB00:0000:0000:1234

Ans — 2001:DB8:0:AB00::1234

Factor increasing IPv6 adoption

Ans — IoT adds millions of network-ready sensors.

Bits in an IPv6 address

Ans — 128 bits.

Co-existence methods

Ans — Translation — converts packets; Dual-Stack — both coexist together; Tunneling — IPv6 inside IPv4.

4 DHCP

Two steps using broadcast to client boot

Ans — DHCP Discover, DHCP Request.

Destination IP for DHCP Discover

Ans — 255.255.255.255.

True about DHCP operation

Ans — Client broadcasts DHCP Discover to find available servers.

Devices typically getting a static IP

Ans — Printers and web servers.

Message to accept offered IP

Ans — DHCP Request (client's broadcast reply to the server's offer).

CISCO NETWORKING BASICS

MODULE 12 — GATEWAYS TO OTHER NETWORKS

🔗 Networking Fundamentals 🔗 Cisco Skills for All ➡ Handwritten Notes → Digital

COURSE Networking Basics	MODULE 12 of Course	STARTED 17 / 08 / 2026	FORMAT Notes + Quiz + Test
-----------------------------	------------------------	---------------------------	-------------------------------

0 Module Introduction

Module 12 explains how a network's **default gateway** connects it to the outside world, and how **NAT** (Network Address Translation) lets private, internal IP addresses talk to the public internet.

1 Network Boundaries & the Default Gateway

- Every host needs a **default gateway** — the IP of the router interface on its own network.
- The gateway address can be set manually or given automatically via DHCP.
- A router (wireless router) acts as the DHCP server for local/internal hosts (wired or wireless).
- A router also acts as a DHCP client toward the ISP — it gets a public/external IP for its internet-facing interface.
- A router is the boundary between the internal (local) network and the external (internet) network.

Rule of same vs. different network:

Scenario	Characteristics
Same network	Different MAC addresses, different IP addresses, but the same default gateway.
Different network	Different MAC addresses, different IP addresses, and a different default gateway.

Diagram — network boundary:



2 NAT Operation

- Router gets a public IP from the ISP → gives private IPs to internal clients.
- **NAT** (Network Address Translation) = converts private IPs ↔ public IPs.
- Outgoing private (local) source IP → translated to a public (global) IP.
- Incoming process is reversed.
- One public IP can represent many internal private IPs (via NAT).
- Only translated when traffic is going to another network (NAT is not applied for local traffic).
- Translation happens at the gateway, replacing the sender's private IP with the router's public IP.

3 Module Test — Questions & Answers

Packet to a destination in the same LAN

Sent directly to the destination host.

Router device performing NAT in a corporate environment

Router (gateway device).

Default gateway characteristic

Logical address of the router interface on the same network as the host.

Purpose of default gateway

To identify the device that allows local network computers to communicate with devices on other networks.

Impact of a wrong default gateway

The host is unable to communicate with hosts on remote networks.

Three types of private IP addresses

10.0.0.0, 172.16.0.0, 192.168.0.0.

Purpose of NAT

Translating private IP addresses to a public, registered IP address.

Primary advantage of NAT

Allows a large group of users to share one or more public IP addresses.

3 settings a PC needs for internet communication

IP address, subnet mask, default gateway address.

Private IP address is a type of

Home router default DHCP address.

Boundary device between home network and internet

Wireless router.

CISCO NETWORKING BASICS

MODULE 13 — THE ARP (ADDRESS RESOLUTION PROTOCOL) PROCESS

Networking Fundamentals

Cisco Skills for All

Handwritten Notes → Digital

COURSE

Networking Basics

MODULE

13 of Course

STARTED

18 / 08 / 2026

FORMAT

Notes + Quiz + Test

0 Module Introduction

Module 13 explains **ARP** — how a host that knows another device's IP address, but not its MAC address, discovers it. It also covers how frames are addressed for local vs. remote destinations, and how broadcasts are contained within a network.

1 MAC Address vs. IP Address

A host may know a destination's IP but not its MAC → it needs **address resolution**.

Two addresses per device on an Ethernet LAN:

Address	Scope
Physical (MAC)	NIC to NIC, on the same network.
Logical (IP)	Source to destination, same or remote network.

- If the destination IP is on a **remote** network → destination MAC = the default gateway's MAC (router interface).
- **Router process:** de-encapsulates the frame → checks the IP → finds the next hop → re-encapsulates in a new frame for the outgoing link.
- If the next hop is the final destination → destination MAC = that device's own NIC.

1. For a frame to a device on the **same** network, the destination MAC is that device's own MAC.
2. For a frame to a device on a **remote** network, the destination MAC is the local router interface's MAC.
3. For both IPv4 and IPv6, ARP (IPv4) and ND - Neighbor Discovery (IPv6) are the protocols used to find a MAC address from an IP.

2 ARP 3-Step Process & Broadcast Containment

Diagram — same-network frame addressing:



Destination MAC = 55-55-55, Source MAC = aa-aa-aa, Source IP = 192.168.10.10, Destination IP = 192.168.10.11.

Broadcast containment:

- Broadcast MAC = 48-bit, all ones = FFFF.FFFF.FFFF in hex.
- Switches forward broadcasts to every host in the same local network → a LAN = one broadcast domain.
- Too many hosts in one broadcast domain → excessive traffic → performance drops.
- Routers divide networks into multiple broadcast domains.
- A NIC only accepts a frame if it's the broadcast MAC or its own MAC.

ARP 3-step process (when only IP is known):

- a. Sender broadcasts a frame containing the target's IPv4 address.
- b. All hosts check the IP; the match sends back its MAC.
- c. Sender stores MAC + IP in its ARP table.

IPv6 equivalent = Neighbour Discovery (ND).

3 In-Chapter Notes — Quick Q&A

Destination MAC of an Ethernet broadcast in hex

FFFF.FFFF.FFFF.

When a switch receives a broadcast frame

It forwards it out all ports except the incoming port.

Host A knows Host B's IP but not MAC

Sends an ARP request.

An ARP request is sent out as

A broadcast, so all devices on the same network receive it.

Host B replies to an ARP request if

The IP address in the request matches its own IP address.

What Host A learns from Host B's ARP reply

Host B's MAC address.

4 Module Test — Questions & Answers

Function of the ARP protocol

Resolving an IPv4 address to a MAC address.

Destination address used in an ARP request frame

FFFF.FFFF.FFFF.

Treatment of ARP requests on the local link

Received and processed by every device on the local network (since ARP requests are broadcasts).

Info examined by a Layer-2 device to forward data

Destination MAC address.

Two functions of MAC addresses

To allow transfer of frames from source to destination, and to uniquely identify a node on a network.

PC1 sends ARP request for PC2

PC2 will send an ARP reply with the PC2 MAC address.

Addresses mapped by ARP

IPv4 address to a destination MAC address.

Sw1 floods frame FFFF.FFFF.FFFF

Sw1 floods out all ports except inbound; Sw2 also floods it; processed by H2, H3, H4.

ARP reply for Host A (different network)

From the default gateway = RTA's Fa0/0 MAC (e.g. 00:0D:00:B4:12:F3).

Characteristic of a MAC address

They are the physical address of the NIC or interface.

Two characteristics of MAC address (choose two)

Physical address assigned to the NIC, and identifies source/destination in the Layer 2 header.

CISCO NETWORKING BASICS

MODULE 14 — ROUTING BETWEEN NETWORKS

Networking Fundamentals Cisco Skills for All Handwritten Notes → Digital

COURSE: Networking Basics MODULE: 14 of Course STARTED: 19 / 08 / 2026 FORMAT: Notes + Quiz + Test

0 Module Introduction

Module 14 explains why networks get split into multiple smaller networks, how a router selects the best path in its **routing table**, and how an IP packet stays intact end-to-end while its Layer 2 (Ethernet) header is rewritten at every hop.

1 The Need for Routing

The need for routing:

- Networks grow → are often split into multiple access-layer networks, based on:

Reason	Description
Broadcast containment	Distribution layer routers limit broadcasts to their local network.
Security requirements	Separate/protect groups with confidential info.
Physical location	Interconnect networks across different geographic sites.
Logical grouping	Group users (e.g. departments) with common needs/resource access.

- The distribution layer connects these local networks and controls traffic between them; keeps local traffic local.
- **Router** = a Layer 3 device connecting multiple IP networks.
- Switches forward by Layer 2 MAC; routers forward by Layer 3 IP.
- If the source and destination network portions of an IP don't match → a router must forward the message.

Diagram — IP packet encapsulated in an Ethernet frame:

Dest. MAC	Source MAC	Source IP	Dest. IP	Data	Trailer
BB:BB:BB:BB:BB:BB	AA:AA:AA:AA:AA:AA	10.0.0.1	192.168.1.5	...	...

A switch examines the MAC address; a router examines the IP address. In IP 192.168.1.5, the network portion and host portion split according to the subnet mask.

2 Multi-Department Network Diagram

Diagram — multi-department network: Internet → Router → Switch 1 (Network Management) → R1 → Switch 2 (Accounting Dept) → R2 → Switch 3 (Sales Dept), each with PCs, phones, laptops, printers, servers, and access points connected.

To maintain smaller broadcast domains and increase network security are the two reasons to divide a network into smaller networks. Routers are the device used to divide a network into smaller networks.

Routing is a process to determine the best path to a destination.

3 The Routing Table

1. Each router port/interface connects to a different local network.
2. Router keeps a table of all locally connected networks + their interfaces.
3. **Process:** router decodes frame → reads destination IP → matches network portion to routing table → if found, re-encapsulates & forwards out the correct interface = routing.
4. Router forwards to either: a directly connected network (host is there), or another router along the path.
5. Destination MAC needed to forward: use the MAC of the connected router (learned via ARP).
6. Host learns the router's IP via its default gateway setting = router interface on the host's own network.
7. Routing table entries are added two ways: dynamically (from other routers) or manually (by admin).

Routing table worked example: S1 (192.168.1.0/24, Fa0/1) — Router1 (192.168.1.1 / 192.168.2.1) — S2 (192.168.2.0/24, Fa0/2), with H1 (AA-AA, 192.168.1.10), H2 (BB-BB, 192.168.1.20), H3 (CC-CC, 192.168.2.50), H4 (DD-DD, 192.168.2.60).

Scenario	Dest. MAC	Source MAC	Dest. IPv4	Source IPv4
H1 → H2 (same network)	BB-BB	AA-AA	192.168.1.20	192.168.1.10

H1 → H3 (remote network)	11-11 (Router)	AA-AA	192.168.2.50	192.168.1.10
--------------------------	----------------	-------	--------------	--------------

When the destination is on a remote network, the destination MAC becomes the router's interface MAC — not the final host's MAC. The destination IP always stays the final host's IP.

4 Default Gateways & Building a LAN

Diagram — two-router topology: Fast Ethernet 0/0 (10.1.21.1, .2, .3, .4) — Default Gateway (192.168.1.254) — Fast Ethernet 0/1 (192.168.1.1/1, .2/1, .3/1, .1/1).

Notes:

- Info in an IP packet the router uses to pick the forwarding interface → the destination IP address.
- Host A encapsulates a packet with the default gateway's MAC when Host B is on a different network.
- A default route is the interface through which the router forwards a packet containing a destination IP network address that is not in the routing table.
- A host sends a packet to the default gateway when the destination IP address is on a different network.

Creating a LAN:

- LAN = a local network or group of interconnected local networks under one administrative control.
- Typical traits: uses Ethernet/wireless, supports high data rates.
- Can place all hosts on one network (single broadcast domain, hosts find each other via ARP), or split across multiple networks via a distribution layer device.
- Splitting hosts across networks reduces traffic impact, but needs routing for them to communicate.
- **Trade-off:** routers add complexity and can introduce latency.

5 Module Test — Important Points

Info routers use to forward a packet

Destination IP address.

Impact of an incorrectly configured default gateway

Host can communicate on the local network, but not with remote networks.

Role of a router on a network

Selecting the path to destination networks (routers use destination IP to forward packets to the right network).

Address configured as default gateway on a client

IPv4 address of the router interface connected to the same LAN.

Device that transfers data from local to remote network

Router.

Device required to connect two separate IP networks

Router.

Benefit of adding a router within an IP network

Keeps broadcasts contained within a local network.

R1 receives packet with destination 255.255.255.255

Examines the packet received on Fast Ethernet 0/1 and does not forward it (broadcasts aren't routed).

Router receives frame with broadcast MAC address

It will NOT forward the frame to another network (router checks if it needs to act on it, like ARP/DHCP, otherwise drops it).

Two reasons to install routers to segment a network

To create smaller broadcast domains; to expand the network to a different geographic location.

Table router uses to pick outgoing interface

Routing table.

Router action when no route found for destination

It drops the packet.

CISCO NETWORKING BASICS

📄 CHECKPOINT EXAM — COMMUNICATION BETWEEN NETWORKS

📖 Checkpoint Exam

🔗 Covers Modules 12-14

⇌ Handwritten Notes → Digital

COURSE

Networking Basics

TYPE

Checkpoint Exam

STARTED

20 / 08 / 2026

FORMAT

20 Questions

0 About This Checkpoint

This checkpoint exam covers **Modules 12-14** — gateways, NAT, ARP, and routing between networks.

1 Checkpoint Exam — Questions & Answers

Q1. LAN devices communicate on the internet via

All devices share the single public IPv4 address assigned to the wireless router (via NAT).

Q2. Result if default gateway is misconfigured on a PC

PC can communicate with the same network, not remote networks.

Q3. User can reach same network but not remote; ping to gateway fails

The default gateway address is incorrect.

Q4. Source IP address when a packet from a home network arrives at a web server (NAT)

A registered public IP address assigned to the external router interface.

Q5. Address type shared via NAT for internet access

Registered public IP address.

Q6. Effect of incorrect default gateway on host

Host cannot communicate with hosts in other networks.

Q7. Purpose of ARP in an IPv4 network

To obtain a specific MAC address when an IP address is known.

Q8. Destination address in an ARP request frame

FFFF.FFFF.FFFF.

Q9. Host A needs to reach Host D — devices that receive the ARP request

Only Host B, Host C, and Router R1.

Q10. ARP action when cache has no mapping

Sends an ARP request to the Ethernet broadcast address to discover the MAC of the destination device.

Q11. Protocol used to find MAC of default gateway

ARP.

Q12. Destination address for ARP request when remote host is unknown

The broadcast MAC address (FFFF.FFFF.FFFF) — ARP requests are always sent as broadcasts.

Q13. ARP table in a switch maps

Layer 3 address to a Layer 2 address.

Q14. Encapsulation of a packet as it passes through a network

Layer 2 header is stripped at each router; Layer 2 header is reassembled at each router.

Q15. Router forwards packet Network 1 → Network 2 — what happens

It reassembles the frame with a different MAC address than the original frame (IP addresses stay the same, only the MAC frame changes).

Q16. Portion of network layer address router uses to forward

Network portion.

Q17. Description of default gateway address

IP address of the router interface that connects the PC LAN to the router.

Q18. Info examined when router forwards packet to remote destination

Destination IP address.

Q19. Why router maintains accurate routing table

To determine the best path to reach the destination network.

Q20. Two functions of a router

A router connects multiple IP networks; it determines the best path to send packets.

CISCO NETWORKING BASICS

MODULE 15 — TCP AND UDP

Networking Fundamentals Cisco Skills for All Handwritten Notes → Digital

COURSE Networking Basics	MODULE 15 of Course	STARTED 21 / 08 / 2026	FORMAT Notes + Quiz + Test
-----------------------------	------------------------	---------------------------	-------------------------------

0 Module Introduction

Module 15 compares the two Transport Layer protocols — **TCP** (reliable, connection-oriented) and **UDP** (fast, best-effort) — and covers well-known port numbers that identify which application a segment belongs to.

1 TCP vs. UDP

UDP	TCP
"Best effort" delivery, no acknowledgement of receipt.	Acknowledgements confirm receipt of packets.
Preferred for streaming audio, VoIP (ACKs would slow things down).	Breaks messages into segments, each with a sequence number.
A few lost packets are usually not noticeable.	Tracks segments sent to a specific host from a specific application.
No sequence number.	If no ACK received in time, TCP assumes the segment is lost and retransmits.
—	TCP resends only the lost portion, not the entire message.

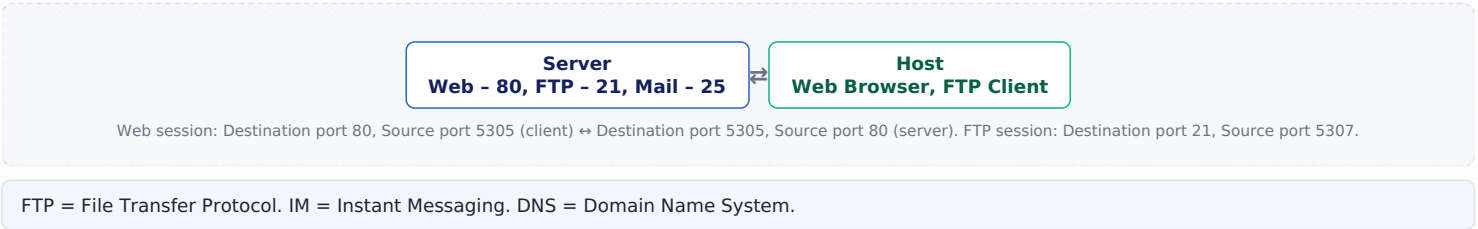
TCP = Transmission Control Protocol — acknowledgements of receipt of packets.

UDP = User Datagram Protocol — no acknowledgements, no sequence number.

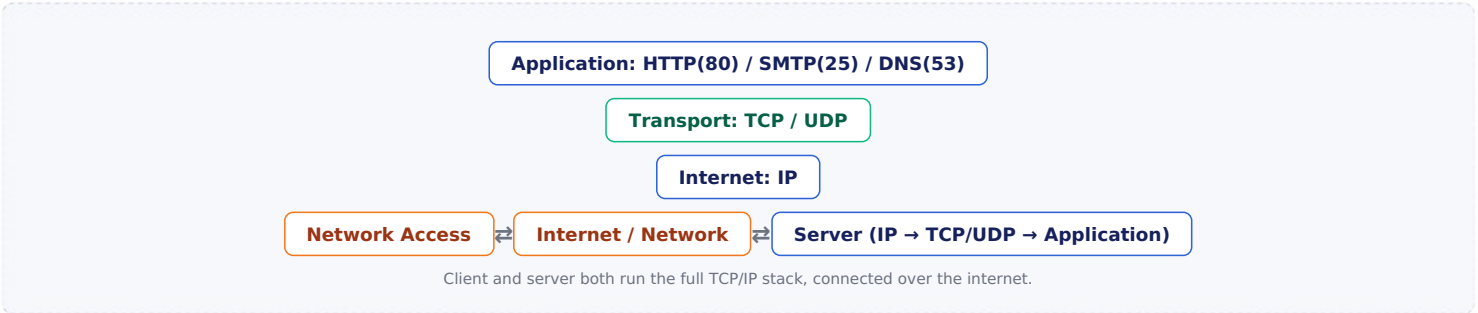
VoIP = Voice over Internet Protocol.

2 Port Communication & the TCP/IP Stack

Diagram — client/server port communication:



TCP/IP diagram (client and server):



Note points:

- The Transport layer ensures packets are sent reliably and resends missing packets in the TCP/IP model.
- TCP keeps track of segments sent to the destination.
- UDP does not use acknowledgements to track receipt of segments.

3 Well-Known Port Numbers

Common well-known port numbers and their associated applications:

Port	Transport	Application Protocol
20	TCP	File Transfer Protocol (FTP) — Data

21	TCP	FTP — Control
22	TCP	Secure Shell (SSH)
23	TCP	Telnet
25	TCP	Simple Mail Transfer Protocol (SMTP)
53	UDP, TCP	Domain Name Service (DNS)
67	UDP	DHCP — Server
68	UDP	DHCP — Client
69	UDP	Trivial File Transfer Protocol (TFTP)
80	TCP	Hypertext Transfer Protocol (HTTP)
110	TCP	Post Office Protocol version 3 (POP3)
143	TCP	Internet Message Access Protocol (IMAP)
161	UDP	Simple Network Management Protocol (SNMP)
443	TCP	HTTP Secure (HTTPS)

- Ports identify protocols/services requested in TCP or UDP messages.
- Every message has a source port and a destination port.
- Ports are assigned/managed by ICANN, range 1–65,535.

Port Range Category	Range	Description
Well-Known Ports	1–1023	Common apps like HTTP, FTP.
Registered Ports	1024–49,151	Used by orgs for specific apps (e.g. IM apps).
Private Ports	49,152–65,535	Dynamically used as source ports by any app.

4 Ports, Sockets & netstat

- Source port is dynamically generated to track separate conversations.
- Destination port tells the server which service is being requested.
- A server can offer multiple services at once (e.g. HTTP on 80, FTP on 21).
- Unexplained TCP connections can be a security threat.
- The netstat command lists protocols in use, local/foreign address & port numbers, and connection state.

Notes:

- Host 10.1.1.10 requests web services from Server 10.1.1.254 — correct socket pair → 10.1.1.10:1099, 10.1.1.254:80.
- Well-known ports port group includes port numbers for FTP, HTTP and TFTP applications.
- The netstat window command shows protocols in use, local/foreign address & port numbers, and connection state.

5 Module Test — Questions & Answers

HTTP protocol operates at which layer?

Application layer of the TCP/IP model.

What UDP provides

Unreliable delivery of segments.

Applications best suited to UDP

Applications that require minimal transmission delay.

Transport layer reassembles messages — example?

Host sends files from phone to computer.

Which protocol does the Transport layer of TCP/IP operate?

TCP.

Port number's role at the transport layer

Identifies a target application.

Well-known port type

Assigned by IANA to commonly used services.

Source port number's role in TCP communication

To keep track of multiple conversations between devices.

Advantage of UDP over TCP

UDP communication requires less overhead.

When UDP is preferred over TCP

When an application can tolerate some loss of data during transmission.

Both UDP and TCP use port numbers

Correctly describes data transmission at the transport layer.

CISCO NETWORKING BASICS

MODULE 16 — APPLICATION LAYER SERVICES

Networking Fundamentals

Cisco Skills for All

Handwritten Notes → Digital

COURSE

Networking Basics

MODULE

16 of Course

STARTED

22 / 08 / 2026

FORMAT

Notes + Quiz + Test

0 Module Introduction

Module 16 covers the **application-layer protocols** that power everyday internet services — web browsing, DNS, email, file transfer, and remote access — and how client/server pairs communicate using specific ports.

1 Client/Server Model & URI/URL/URN

- **Server** = a host running software that provides info/services; **client** = software like a browser requesting them (e.g. browser, web server).
- Client sends request, server responds.

Diagram — web client/server exchange:



Source IP: 172.16.10.50:80 → Destination IP: 192.168.10.15:5507 (server reply); reverse for the client's request.

1. DNS lookup — resolves 172.16.10.50.
2. TCP connection.
3. **URI** identifies a network resource; parts = protocol/scheme, hostname, path, filename, fragment.
4. **URN** identifies namespace only; **URL** defines the exact network location of a resource.
5. Common internet services: search, social media, streaming, shopping, email, messaging.
6. Common app-layer protocols: DNS, SSH, SMTP, POP, IMAP, DHCP, HTTP, FTP.

2 URI / URL / URN — Structure

Diagram — instant messaging client/server: both clients can simultaneously send and receive messages, relayed through the network.

Term	Meaning
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
URN	Uniform Resource Name

Components of a URI/URL/URN (example: <https://www.example.com/author/book.html#page1>):

Component	Covers
URL	The full address: scheme + host + path + filename.
URN	host + path (the resource's namespace).
Fragment	The part after "#" (page1) — a specific location within the resource.
URI	The entire string — covers URL, URN, and fragment together.

3 DNS, HTTP/HTTPS & Email Flow

7. DNS resolves hostname to IP addresses using domains (.com, .edu, .net).
8. If the local DNS server has no entry, it queries another DNS server in the domain.
9. Web client uses IP + port 80 to request pages via HTTP.
10. HTTP is not secure (plaintext); HTTPS (port 443) provides secure transport.
11. Web page content is encoded in HTML, which tells the browser how to format/display it.

Diagram — data center services: a switch connects to DNS Server, SSH Server, Email Server, DHCP Server, Web Server, FTP Server.

Diagram — email flow: Sender → Internet → SMTP → Mail Server (ISP A) ↔ SMTP ↔ Mail Server (ISP B) → IMAP or POP3 → Internet → Recipient.

4 FTP, Telnet, SSH & Email Protocols

#	Service	Key Facts
12	FTP	Transfers files between client and server; supports upload, download, delete, rename.
13	FTP	Uses two ports: TCP 21 (control connection) and TCP 20 (data transfer).
14	Telnet	Emulates a text-based terminal session (VTY) using TCP port 23.
15	Telnet	Insecure — sends data as plaintext, easily intercepted.
16	SSH	Provides secure remote login with encrypted, stronger authentication — preferred over Telnet.
17	SMTP	Sends email from client to mail server; uses port 25.
18	POP3	Downloads email to client, doesn't keep messages on server by default; uses port 110.
19	IMAP4	Keeps messages on the server (unless deleted); uses port 143.

20. Text messaging = instant/direct/private/chat messages; real-time internet chat.

21. Internet telephony uses peer-to-peer tech; VoIP converts voice to digital data carried in IP packets.

FTP session diagram:

1. **Control connection:** client opens first connection to the server for control traffic.
2. **Data connection:** client opens second connection for data traffic.
3. **Data transfer:** server transfers data to the client.

Based on commands sent across the control connection, data can be downloaded from the server or uploaded from the client.

5 Notes — Email & File Transfer Clients

- DHCP application protocol automatically configures IP addressing information.
- POP and IMAP application protocols are used by clients to retrieve email messages.
- DNS application protocol resolves web addresses to an IP address.
- Client-to-mail-server send protocol is SMTP.
- IMAP feature: downloads a copy of email, leaves the original on the server.
- A single computer can run multiple types of client software simultaneously (e.g. browser + email client + FTP client at once).
- Web-based email client examples: Microsoft 365, Yahoo, Gmail.
- FTP clients come in two forms: command-line interface (built into Windows, macOS, Linux) and GUI-based with drag-and-drop.

6 Module Test — Questions & Answers

Two characteristics of an FTP connection

Files can be downloaded/uploaded to server; first connection = traffic control, second = file transfer.

What protocols define

How messages are exchanged between source and destination in a network.

HTTP protocol used by

Web servers to serve up a web page.

DHCP / SSH / POP / DNS roles

DHCP = automatically configures IP. SSH = remote access to servers. POP = retrieves email messages. DNS = resolves internet names to IP.

SMTP, POP protocol used in

Sending / receiving emails.

SSH, Telnet applications provide

Virtual terminal access to remote servers.

SSH provides secure communications to access hosts, of using SSH over Telnet

True.

DNS protocol allows

Typing `www.cisco.com` instead of an IP address.

HTTP protocol transfers

Web pages from a server to a client device.

HTTP, HTTPS application layer protocols manage

Exchange between browser and web server.

Port numbers for email protocols

IMAP4 — 143, POP3 — 110, SMTP — 25.

CISCO NETWORKING BASICS

MODULE 17 — NETWORK TESTING UTILITIES (FINAL MODULE)

🔍 Networking Fundamentals 🔗 Cisco Skills for All ➡ Handwritten Notes → Digital 📖 Final Course Module

COURSE	MODULE	STARTED	FORMAT
Networking Basics	17 of 17	23 / 08 / 2026	Notes + Quiz + Test

0 Module Introduction

The final module of Networking Basics! Module 17 covers the everyday **CLI troubleshooting utilities** — `ipconfig`, `ping`, `netstat`, `tracert`, and `nslookup` — that every SOC analyst and network admin reaches for first when diagnosing a connectivity issue.

1 Network Testing Utilities — Overview

Several software utilities help identify network problems; most are OS-provided CLI commands.

Utility	Function
ipconfig	Displays IP configuration info.
ping	Tests connections to other IP hosts.
netstat	Displays network connections.
tracert	Displays the route taken to a destination.
nslookup	Queries the name server for info on a destination domain.

2 ipconfig — Details

- Basic `ipconfig` shows: IP address, subnet mask, default gateway.
- `ipconfig /all` shows additional info: MAC address, default gateway IP, DNS server IPs, DHCP status, DHCP server address, lease info.
- `ipconfig /release` — release current DHCP bindings.
- `ipconfig /renew` — requests fresh config info from the DHCP server.
- A host with faulty/outdated IP config often just needs a renewal to regain connectivity.

3 ping

- Most commonly used network utility.
- Tests whether network devices are reachable over IP.
- Sends an echo request packet to the target IP.
- If the destination receives it, it replies with an echo reply.
- Receiving the echo reply back confirms connectivity to that IP address.

Useful ping switches:

Switch	Function
ping -4	IPv4 only.
ping -6	IPv6 only.
ping -a	Resolve address to hostname.
ping -t	Continue pinging until manually stopped.

4 Module Test — Important Points

ping 10.1.1.1 command is used to check

If there is connectivity with the destination device.

Helpdesk asks user to ping the default gateway — purpose

To test that the host has the capability to reach hosts on other networks.

Info revealed by plain ipconfig (without /all)

Subnet mask, IP address, default gateway.

Command that tests connectivity and gives a response for each packet received

ping — tracert traces the route; ping tests device-to-device connectivity.

Utility to display active network connections on a PC

netstat.

Command to find the MAC address on a host PC

ipconfig /all (displays IP config + gateway address + MAC address).

User can ping www.cisco.com by IP but can't browse to it — cause

The DNS server is not working (DNS resolves the URL to an IP; if IP-ping works but name-ping fails, DNS is the problem).

Technician finds PC has wrong IP from DHCP — fix

ipconfig /renew (requests a fresh IP from DHCP; /release only removes the current IP, doesn't request a new one).

Utility to test connectivity to other IP hosts

ping.

Utility roles recap

nslookup → directly queries the name server for domain info. netstat → displays network connections. tracert → displays the route taken to the destination. ping → tests connections to other IP hosts. ipconfig → displays IP configuration information.

CISCO NETWORKING BASICS

📊 CHECKPOINT EXAM — PROTOCOLS FOR SPECIFIC TASKS

📅 Checkpoint Exam

⚙️ Covers Modules 15-17

🏁 Final Checkpoint of the Course

COURSE

Networking Basics

TYPE

Checkpoint Exam

STARTED

24 / 08 / 2026

FORMAT

17 Questions

0 About This Checkpoint

This is the **final checkpoint exam** of the Networking Basics course, covering Modules 15-17 — TCP/UDP, application layer protocols, and network testing utilities. Only the final course exam remains after this!

1 Checkpoint Exam — Questions & Answers

Q1. Client establishing UDP communication with a server

The client randomly selects a source port number.

Q2. netstat without options displays

A list of all established active TCP connections.

Q3. How TCP resequences data

TCP uses sequence numbers to reassemble and reorder received segments.

Q4. Reassembling packets in correct order (TCP/IP model)

Transport layer.

Q5. Well-known port for secure HTTP (HTTPS) request

443.

Q6. UDP best suited for

Applications that are sensitive to delay.

Q7. TCP segment lost between R1 → R2

The web server won't acknowledge it; PC1's timer expires and PC1 resends the segment.

Q8. Number/set representing a socket

192.168.1.1:80 (IP address + port number combo).

Q9. Three pieces of info identified by a URL

Location of the resource, domain name being accessed, protocol being used.

Q10. Three protocols allowed for email use

SMTP, POP3, IMAP4 (SMTP sends, POP3/IMAP4 receive).

Q11. Web client / HTML / HTTP / Web Server / HTTPS roles

Web client = requests a web page. HTML = used to create a web page. HTTP = protocol commonly used by browser. Web Server = hosts a web page. HTTPS = secure protocol using port 443.

Q12. Function of tracer that differs from ping

tracer shows information of routers in the path (sends 3 pings per hop); ping's main function is testing connectivity between two hosts.

Q13. PC has APIPA address (169.254.33.16) — command to request a new IP

ipconfig /release, then a fresh lease (ipconfig /renew).

Q14. Utility roles recap

ipconfig → display IP config info. ping → test connections to other IP hosts. nslookup → queries name server for domain info. netstat → displays network connections. tracert → display route taken to destination.

Q15. Statement describing the ping command

Tests end-to-end connectivity between two devices.

Q16. Technician has only an endpoint's IP, needs to find the faulty node in the path

tracert.

Q17. ping switches recap

ping -4 → IPv4 only. ping -6 → IPv6 only. ping -a → resolve address to hostname. ping -t → continue pinging until manually stopped.

✓ Course Progress

🎉 **All 17 modules + every checkpoint exam of Cisco Networking Basics are now digitized!** 🎉

Only the Final Course Exam remains.

CISCO NETWORKING BASICS

📖 COURSE FINAL EXAM — REVIEW NOTES

📅 Final Exam

📊 Score: 98% (Passed)

🏆 Course Complete

COURSE	TYPE	STARTED	SCORE
Networking Basics	Final Course Exam	25 / 08 / 2026	98% — Passed

0 About the Final Exam

The Networking Basics course final exam — **passed with a 98% score**. This review covers the questions worth revisiting; several easy questions were intentionally skipped since the concepts were already clear from earlier module notes.

1 Wireless & Mobile Networking

Q2. Tourist needs laptop internet via smartphone (3G/4G).

Ans — Enable tethering and create a hotspot.

Q3. Three steps to manually connect Android/iOS to a secured wireless network.

Ans — Choose the correct security type; enter the network SSID; input the authentication password.

Q4. Wireless RF band used by IEEE 802.11b/g devices.

Ans — 2.4 GHz.

Q5. Low-cost internet connectivity method for a mobile device at a hotel.

Ans — Wi-Fi.

Q6. Disadvantage of a peer-to-peer network model.

Ans — Lack of centralized administration.

Q7. Statement describing powerline networking technology.

Ans — A device connects to an existing home LAN using an adapter and an existing electrical outlet.

Q8. Three characteristics of a peer-to-peer network.

Ans — Less cost to implement; easy to create; lacks centralized administration.

Q9. Two methods typically used on a mobile device for internet connectivity.

Ans — Cellular, Wi-Fi.

Q10. Action needed to prevent neighbors from discovering a wireless network.

Ans — Disable SSID broadcast.

2 Network Media & Switching

Q11. Advantage of fiber-optic cabling compared to copper cabling.

Ans — Fiber-optic cabling is completely immune to EMI and RFI.

Q12. Scenario suitable for deploying twisted-pair cables.

Ans — To connect PC workstations in an office.

Q13. Cisco LAN switch action when destination MAC isn't in the MAC address table.

Ans — Forward the frame out all ports except the port where the frame was received.

Q14. Data encoding technology used in copper cables.

Ans — Electrical pulses.

Q19. Two criteria used to help select a network medium.

Ans — The environment where the medium is installed; the distance the medium can carry a signal.

Q20. Information added to the switch table from incoming frames.

Ans — Source MAC address and incoming port number.

3 OSI & TCP/IP Models

Q15. Two OSI model layers with the same functionality as two TCP/IP model layers.

Ans — Transport, network.

Q16. TCP/IP layer responsible for routing messages through an internetwork.

Ans — Internet layer.

Q17. Network model type that describes required functions without specifying exact protocol behavior.

Ans — Reference model.

Q18. OSI layer at which a logical address is added during encapsulation.

Ans — Network layer.

4 IPv4 / IPv6 Addressing

Q21. Two groupings used to create a college's IP addressing scheme (device-based ranges).

Ans — Device type, geographic location.

Q25. Bits represented by each group of four hexadecimal values in an IPv6 address.

Ans — 16 bits.

Q26. How a host determines if a packet's destination is local or remote.

Ans — It uses the subnet mask to compare the source and destination IP address.

Q28. Layer 2 and Layer 3 destination addresses in a DHCP request message.

Ans — FF-FF-FF-FF-FF-FF and 255.255.255.255.

Q30. Purpose for which IPv4 addresses are utilized.

Ans — To uniquely identify a device on an IP network.

5 Gateways, NAT & Routing

Q34. How the global address is assigned for NAT on a home wireless router.

Ans — The wireless router acts as a DHCP client to receive global addressing from the ISP.

Q36. Two scenarios where a wireless router performs NAT.

Ans — When a host sends packets to the ISP; when a host sends packets to a remote site (manufacturer) to request a device manual.

Q39. Best device to reduce heavy broadcast traffic on a multi-floor LAN.

Ans — Router.

6 Transport Layer & Troubleshooting

Q41. Command used to display network connections on a host computer.

Ans — netstat.

Q43. Possible cause for a failed ping to the default gateway (while ping to an outside host succeeds).

Ans — Security rules are applied to the default gateway device, preventing it from processing ping requests.

Q44. How port numbers are used in the TCP/IP encapsulation process.

Ans — If multiple conversations use the same service, the source port number tracks the separate conversations.

Q47. CLI command to help locate a network problem (local ping OK, internet ping fails).

Ans — tracer.

Q49. Format that analog voice signals are encapsulated in for use on a computer network.

Ans — IP packets.

7 Application Layer — Protocol Matching

Q50 — Match protocol/function to description (website visit scenario):

Description	Protocol / Function
A user types a URL and a page loads in the browser.	Web service
A user requests a website by typing its name.	Domain Name System (DNS)
A workstation gets its IP address automatically.	Dynamic Host Configuration (DHCP)
A service that lets workers send messages over the internet.	Email
A network technician securely accesses a device remotely.	SSH
A user uploads new files to their website.	File Transfer (FTP)

Exam Result

 **Networking Basics — Course Final Exam: 98% — PASSED** 

This closes out the full 17-module Networking Basics course, including all checkpoint exams. Certificates and badge on the following page.

CISCO NETWORKING BASICS

CERTIFICATES & ACHIEVEMENTS

Cisco Networking Academy

Verified Credentials

Course Complete

COURSE

Networking Basics

LEARNER

Shivam Kumar

PROVIDER

Cisco Networking Academy

STATUS

Complete

Course Completion Certificate

Cisco
Networking
Academy

This certificate is awarded to

Shivam Kumar

for successfully completing

Networking Basics

offered by Networking Academy
through the Cisco Networking Academy program.

Lynn Bloomer

Lynn Bloomer
Director
Cisco Networking Academy



05 Sep 2026
Completion Date

Cert ID: aa32b24e-a831-4853-a1bd-1cec7bac8838

Awarded to Shivam Kumar for successfully completing **Networking Basics**, offered by Networking Academy through the Cisco Networking Academy program. Completion Date: 05 Sep 2026.

Cisco Packet Tracer Certificate

Cisco
Networking
Academy

This certificate is awarded to

Shivam Kumar

for successfully completing

Getting Started with Cisco Packet Tracer

offered by Networking Academy
through the Cisco Networking Academy program.

Lynn Bloomer

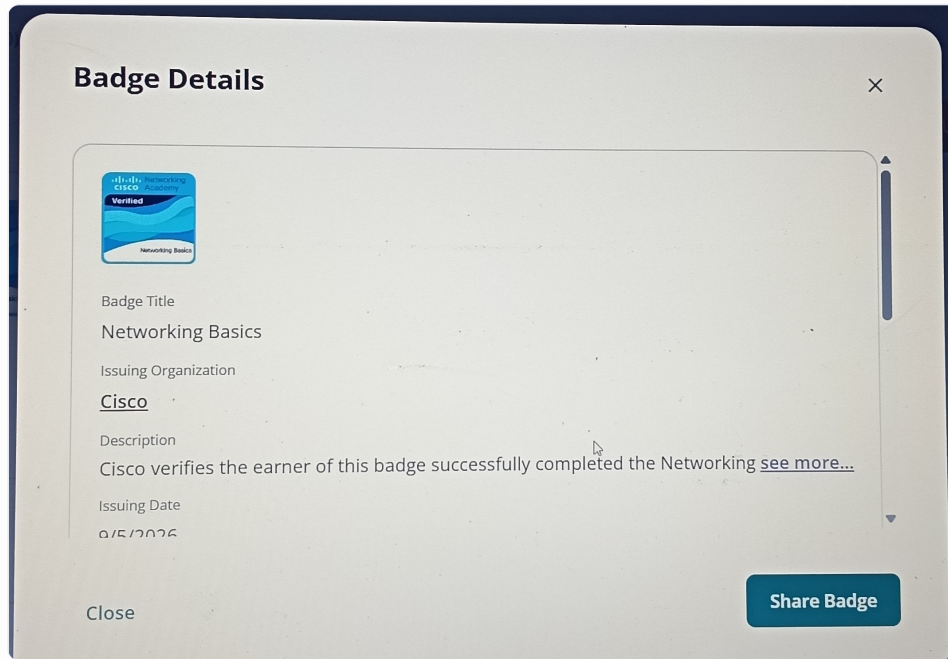
Lynn Bloomer
Director
Cisco Networking Academy



01 Sep 2026
Completion Date

Cert ID: 93a617ad-0deb-4515-8d3e-30dac9fb6e1

✓ Credly Verified Badge



Networking Basics — Verified badge issued by Cisco on Credly. Issuing Organization: Cisco. Issued: 5 Sep 2026.

✓ Summary

🎖 **Networking Basics — fully complete: 17 modules, 5 checkpoint exams, final exam (98%), Cisco Packet Tracer certificate, course certificate, and a verified Credly badge.** 🎖

Phase 1 (Semester 2) networking foundation of the SOC L1 roadmap — complete.